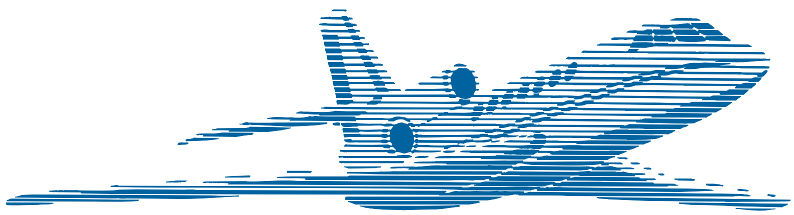
FlightSafety international

FALCON 50

EMERGENCY/ABNORMAL PROCEDURES



Revision 9.4

PILOT CHECKLIST

INSERT LATEST REVISED PAGES, DESTROY SUPERSEDED PAGES
LIST OF EFFECTIVE PAGES

Dates of issue for original and changed pages are:

Revision	2	January 1999
Revision	3	April 2000
Revision	4	July 2000
Revision	4.1	March 2002
Revision	5	January 2003
Revision	6	April 2004
Revision	7	January 2010
Revision	8	March 2010
Revision	8.1	June 2015
Revision	8.2	July 2015
Revision	8.3	February 2018
Revision	8.4	April 2019
Revision	8.5	August 2019
Revision	8.6	January 2020
Revision	9.0	January 2022
Revision	9.1	February 2022
Revision	9.2	August 2023
Revision	9.3	October 2023
Revision	9.4	November 2023

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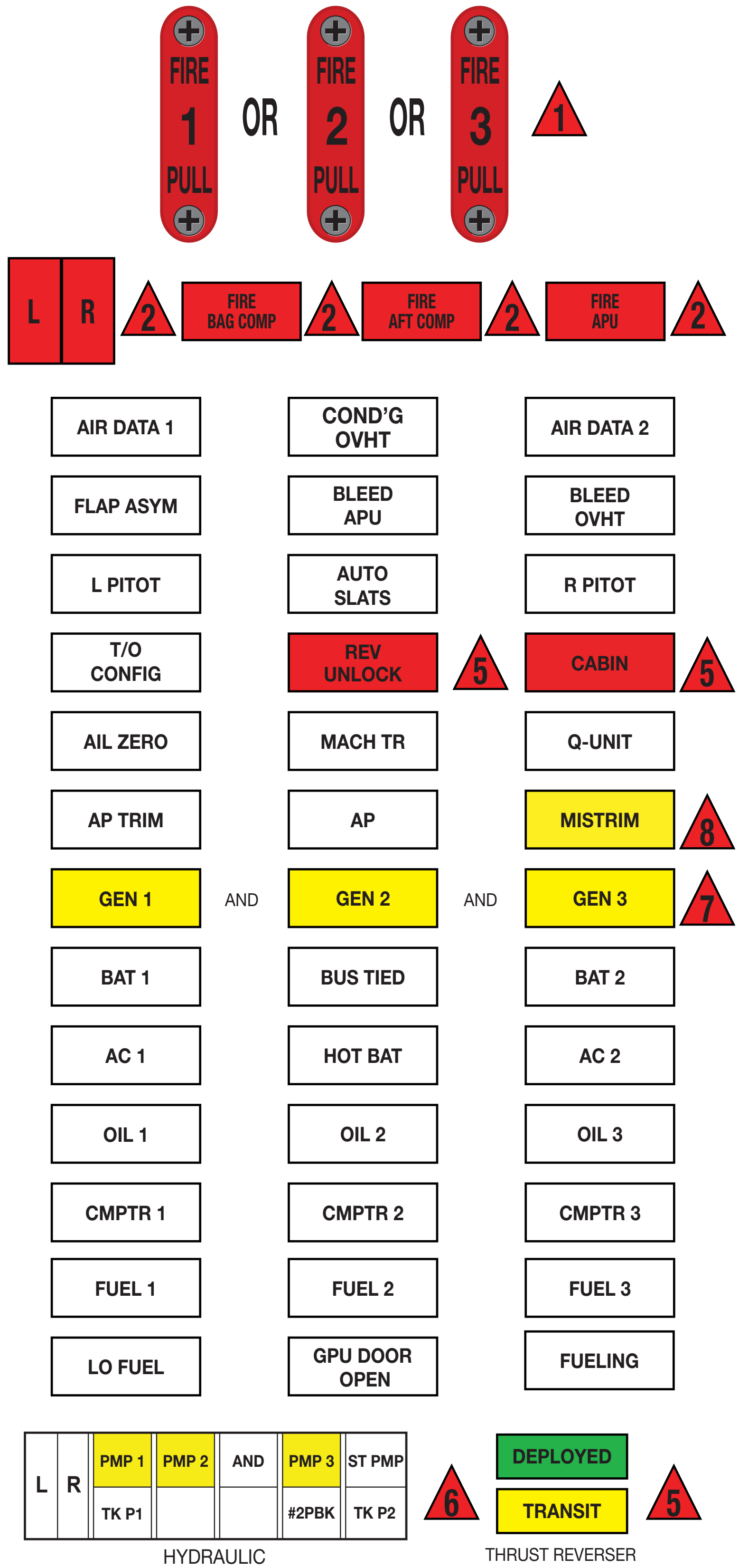
These are suggested procedures only and in no way supersede current procedures outlined in the FAA-approved Flight Manual and any revisions thereto. In the case of conflict, the Flight Manual takes precedence.

EMERGENCY PROCEDURES

Compliance with the order prescribed for application of these procedures is recommended.

PHASE I	These items require immediate action for the safety of flight. It is recommended to perform them from memory.
PHASE II	These items shall be completed only after the PHASE I items have been completed by the checklist.
PHASE III	These items shall be completed as soon as time permits.

AURAL WARNINGS shall be identified before silencing. Once the mafunction has been identified, silencing the aural warning will enable better coordination in performing the emergency procedure.



FOR TRAINING PURPOSES ONLY

EMERGENCY PROCEDURES

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1

FIRE AND SMOKE

ENGINE FIRE IN FLIGHT

+ AURAL WARNING

Phase I

1. Power Lever...CUTOFF
2. FIRE PULL handle .... PULL
3. Airspeed.. BELOW 250 KIAS
4. Fire Extinguisher Switch Engine ....POSITION 1

If the fire persists:

5. Fire Extinguisher Switch ....POSITION 2

Phase II

Complete shutdown of engine:

6. Booster Pump... OFF
7. Generator Switch.. OFF

*NOTE

In icing conditions, the #1 or #2 engine anti-ice system should be operated even with the #1 or #2 engine shutdown.

*8. Engine Anti-ice.. OFF

9. HP Bleed (and PRV in the case of #2 Engine) .... OFF

If #3 engine is shutdown:

10. Bus Tie Switch.. TIED
11. Hydraulic Standby Pump................................... ON (AS REQUIRED)
12. Land as soon as possible.
13. Fuel Management .... as required

OR

One Engine Inoperative Approach/Landing/Go-Around Procedures:

• Yellow Tab 6, page A-12

Approach And Landing With Two Engines Inoperative:

• Red Tab 10, pages E-24 and E-25

ENGINE FIRE ON GROUND

+ AURAL WARNING

Phase I 1

1. Power Lever.. IDLE
2. Airplane..STOP
3. Parking Brake ....FIRST DETENT
4. Power lever...CUT OFF
5. FIRE PULL handle .... PULLED
6. Fire Extinguisher ENG...POSITION 1
7. Fire Extinguisher ENG...POSITION 2
8. EMERGENCY EVACUATION ....Do

Consult Red Tab 15 page E-35.

TAIL PIPE FIRE

There will be no warning or alarm in the cockpit (the turbine area is designed to withstand very high temperatures);

Detection and reporting mainly rely on ground crew or ATC.

Phase I

1. Power Lever..CUT OFF
2. Start selector switch.......................................MOTOR START STOP

   De–activates the igniters and enables motoring to ventilate the . engine core

3. Start Pushbutton...................Held DEPRESSED FOR 15 SECONDS

Phase II

If the fire is reported out

4. Start selector switch...GRD START

If the fire cannot be contained before the end of the motoring sequence:

5. EMERGENCY EVACUATION ....DO

Consult Red Tab 15 page E-35.

SMOKE IN THE BAGGAGE COMPARTMENT

FIRE BAG COMP + AURAL WARNING

Phase I

2 1. Bag Air Switch .... OFF
2. Bag Compartment Fire Extinguisher .... POSITION 1
3. Omega (if equip) .... OFF

Phase II

If INS malfunction(s) observed:

4. INS .... OFF
5. LAND AS SOON AS POSSIBLE.

NOTE

Aircraft w/Aft Avionics Cooling, Crew Bleed Air Switch – OFF.

APU FIRE

FIRE APU + AURAL WARNING

Phase I

1. APU Master .... PUSHED
2. APU Fire Extinguisher Switch .... POSITION 1

Phase II

3. APU Bleed .... OFF
4. APU Generator Switch .... OFF

If the fire persists:

5. #2 Engine Fire Handle .... PULL

Complete shutdown of #2 engine:

6. Power Lever .... CUTOFF
7. Booster Pump .... OFF
8. Generator Switch .... OFF
9. Engine Anti-ice .... OFF
10. HP Bleed Switch .... OFF
11. PRV Switch .... OFF

Evacuate the aircraft.

Consult Emergency Evacuation procedure Red Tab 15 page E-35.

AFT COMPARTMENT FIRE

FIRE AFT COMP + AURAL WARNING

NOTE

A pause should be made after each operation to check resulting effects.

Phase I

1. HP Bleed and PRV (All Switches) .... OFF
2. #2 Engine Anti-ice Switch... OFF
3. Battery Switches.. BOTH OFF
4. Hydraulic Standby Pump Switch.. OFF

If the fire persists:

5. AFT Compartment Fire Extinguisher Switch ....POSITION 1

Phase II

If the fire persists:

6. ITT (All Three Engines) ....CROSS-CHECK
7. HOT Engine .... IDLE
8. LAND AS SOON AS POSSIBLE.

Once the fire is extinguished, all circuits in operating condition may be re-activated to continue flight to nearest landing airport.

2

WHEEL WELL OVERHEAT

L AND/OR R + AURAL WARNING

1. Airspeed... 190 KIAS OR LESS
2. Landing Gear ....DOWN

Keep the landing gear extended until the WHEEL L AND/OR R lights go out, but not less than 10 minutes.

CAUTION

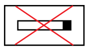
The overheat condition may have caused the tires to deflate. Make a shallow final approach and as soft a landing as possible.

ELECTRICAL SMOKE OR FIRE

Indications: Smoke and unusual odors

Phase 1

1. Crew Oxygen Masks.................... DONNED—100% + EMERGENCY
2. Smoke Goggles .... DONNED—Vent valve open
3. Microphone Selector .... MASK—Tested

3 4.  Light Pushbutton .... ON

ONLY IF NO FLAME IN CABIN:

5. PASSENGER OXYGEN Controller .... OVERRIDE
6. Passenger Masks .... DONNED—Checked

Phase II

7. Crew Air Gaspers.. OPEN
8. E. BATT Switch .... OFF
9. #2 Battery Switch .... OFF
10. #3 Generator Switch.. OFF
11. Bus Tie Switch.. FLT NORMAL
12. Crew Temp Controller .... MANUAL/COLD

Continued on Next Page

OR

If Smoke Persists:

13. #2 Battery Switch .... ON
14. #3 Generator Switch......... ON
15. #1 Battery Switch ....OFF
16. #1 Generator Switch........OFF
17. #2 Generator Switch........OFF
18. Passenger Temp Controller ..... MANUAL/COLD
19. Cabin Pressure Selector Switch....... MANUAL
20. Manual Pressurization Knob ....UP
21. Cabin Altitude ....MONITOR

If Smoke Continues:

Apply the Smoke Removal checklist; Tab 4, page E-9.

If the fire is NOT verified to be out:

22. Land as soon as possible.

If Smoke Disappears:

Keep the faulty equipment off.

If Smoke Disappears:

Keep the faulty equipment off.

If the fire IS verified to be out:

22. Continue the flight. (Land as soon as practical)

3

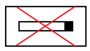
See Key Bus Item/Listing; Blue Tab 3, page QR-4.

4

AIR CONDITIONING SMOKE

Indication: Smoke at the Air Conditioning outlets

Phase I

1. Crew Oxygen Masks.................... DONNED—100% + EMERGENCY
2. Smoke Goggles .... DONNED—Vent valve open
3. Microphone Selector .... MASK—Tested
4.  Light Pushbutton .... ON
5. PASSENGER OXYGEN Controller .... OVERRIDE
6. Passenger Masks .... DONNED—Checked

Phase II

7. Crew Air Gaspers.. OPEN
8. Bleed-air Isolation Knob .... ISOLATION
9. Crew Bleed-air Switch .... OFF
10. PRV Switch .... OFF
11. Temperature Controllers (both) .... MANUAL/COLD

OR

If Smoke Persists:

12. Crew Bleed
    Air Switch......................... ON
13. Cabin Bleed
    Air Switch...................... OFF

If Smoke Continues:

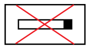
14. Crew Bleed
    Air Switch.......................OFF
    Apply the Smoke Removal checklist;
    Tab 4, page E-9.
15. Descent.....................Initiate

If Smoke Disappears:

Continue the flight with the faulty BLEED AIR system isolated.

SMOKE REMOVAL

Phase I

1. Crew Oxygen Masks.................... DONNED—100% + EMERGENCY
2. Smoke Goggles .... DONNED—Vent valve open
3. Microphone Selector .... MASK—Tested
4.  Light Pushbutton .... ON
5. Crew air gaspers.. OPEN

ONLY IF NO FLAME IN CABIN:

6. PASSENGER OXYGEN Controller .... OVERRIDE
7. Passenger Masks .... DONNED—Checked
8. Passenger cabin air gaspers .... OPEN

Phase II 4

9. Temperature Controllers .... MANUAL/COLD

   Descend to below 12,000 feet, or to a safe altitude for the route flown.

CAUTION

The following procedure must not be applied if flames are present in the cabin or cockpit.

At 12,000 feet or below:

10. Cabin Pressurization Selector ....(A BUS PWR) DUMP
11. Manual Pressurization Knob.. UP

At 180 knots or less:

12. DV Window .... OPEN

Phase III

If smoke persists:

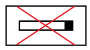
13. LAND AS SOON AS POSSIBLE.

PRESSURIZATION

RAPID DEPRESSURIZATION

CABIN + AURAL WARNING

Phase I

1. Crew Oxygen Masks.................... DONNED—100% + EMERGENCY
2. Microphone Selector .... MASK—Tested
3. FASTEN BELTS and  Light Pushbuttons .... ON
4. PASSENGER OXYGEN Controller .... OVERRIDE
   a. Passenger Masks .... DONNED—Checked
5. Emergency descent .... INITIATED

5

EMERGENCY DESCENT

CAUTION

This procedure assumes structural integrity of the aircraft. If structural integrity is questioned, limit airspeed to lowest practical value, and avoid high maneuvering loads.

Phase

1. Autopilot ....DISENGAGE
2. Power Levers .... IDLE
3. Airbrakes..POSITION #2
4. Descent Airspeed .... V_{MO}/M_{MO} (SMOOTH AIR)
5. Transponder..MAYDAY CODE “7700”

THRUST REVERSER DEPLOYMENT IN FLIGHT

REV UNLOCK AND POSSIBLY DEPLOYED / TRANSIT

Pitch down moment. Abnormal noise and buffeting.

Phase I

1. #2 Power Lever .... IDLE
2. T/R Emergency Stow Switch .... STOW
3. Airspeed .... 230 KIAS OR LESS

Phase II

If the thrust reverser stows, continue the flight with the EMERG/STOW switch in the STOW position.

If the thrust reverser remains deployed:

4. T/R Emergency Stow Switch .... MAINTAIN IN STOW POSITION

With thrust reverser deployed:

5. Land as soon as possible.

NOTE

The Actuator ensures stowing of the Reverser up to 180 KIAS.

NOTE

The drag resulting from an idle #2 engine with the thrust reverser deployed adversely affects the performance characteristics. The engine should therefore be shutdown whenever necessary.

Engine Shutdown:

1. Power Lever (Idle for 1 minute if possible) .... CUTOFF
2. Booster Pump .... OFF
3. Generator Switch .... OFF

NOTE

In icing conditions, operate the #2 engine anti-ice even though the #2 engine is inoperative.

4 Engine Anti-ice .... OFF
5. HP Bleed .... OFF
6. PRV .... OFF
7. Fire Handle .... PULL

OR

One Engine Inoperative Approach/Landing/Go-Around Procedures:

• Yellow Tab 6, page A-12

Approach And Landing With Two Engines Inoperative:

• Red Tab 10, pages E-24 and E-25

5

LOSS OF BOTH HYDRAULIC SYSTEMS

PMP 1 **PMP 2** **PMP 3** AND POSSIBLY **Q UNIT**

WITH BOTH INDICATORS HYDRAULIC PRESSURE DROP AND FLUID LEVELS IN THE RED

Phase I

1. Autopilot and Yaw Damper .... OFF
2. Airspeed .... 260 KIAS/0.76 M OR LESS

CAUTION

The hydraulic power off condition results in greater pilot forces, and landing requires increased caution because directional control is available mainly by rudder and differential forward thrust.

Phase II

Avoid high pitch attitudes and zones of turbulence.

Landing:

3. Slat/Flap Handle ....LEFT IN POSITION

6

NOTE

Without pressure in the two hydraulic systems the Slat/flap handle must not be moved.

4. Approach Speed... BUGS SET

   Clean..VREF+ 30 KNOTS

NOTE

In the situation where the high lift devices are already extended, observe the following approach speeds:

Slats Only.. VREF + 20 KNOTS

Slats + Flaps 20 .... VREF + 15 KNOTS

Slats + Flaps 48 .... VREF + 10 KNOTS

5. Landing Gear .... FREE FALL EXTENSION

Consult Landing Gear Emergency Extension, Abnormal section; Tab 11, page A-22.

6. Final Approach Vertical Speed .... 300 FT/MIN

NOTE

With no antiskid, add 30% to L.D./L.F.L. With no slats/flaps, add 3000/5000 feet. With no airbrakes, add 600/1000 feet. Check AFM (P.5.50.8) whether speed VREF +30 KT is less than VMBE. (See Tab 2, QR-3: Landing Distance/ Landing Field Length Additives.)

After touchdown:

7. Thrust Reverse... FULL POWER
8. Parking Brake .... *INTERMEDIATE DETENT

   *Be cautious and avoid cycling pressure on and off.

UNRELIABLE AIRSPEEDS AT HIGH ALTITUDE

WARNING - Frozen or abnormal pilot and copilot IAS / MI indications and possibly:

- **AIRSPEED INDICATORS PERFORM LIKE ALTIMETERS** (airspeed decreasing in descent and increasing in climb),
- Illumination of one or both following lights:
  Q UNIT **AUTO SLATS**
- VMO / MMO audio warning sounds,
- **IAS** miscompare flag on EADI (if any),
- AP disengagement,
- Disagreement with stand-by IAS / MI indications,
- In cruise / level flight: unusual pitch trim activity.

PHASE I

1. AP .... DISENGAGED
2. YD .... DISENGAGED
3. Avoid large displacements and rapid movements of control surfaces.
4. Fly wings level.
5. Stabilize airplane altitude using, if necessary, the stand-by instrument altitude indication.
   - Pitch attitude.................................... Between 1° and 4° nose up
   - Engine power .... Smoothly fully forward

WARNING

INAPPROPRIATE FLIGHT DIRECTOR GUIDANCE MAY BE ACTIVATED.

DO NOT FOLLOW CORRESPONDING FD.

CAUTION

Stall aural warning remains reliable.

Continued on Next Page

PHASE II

1. Do not apply SLAT MONITORING SYSTEM procedure (Yellow Tab 10).

CAUTION

Do not re-engage AP or YD before pitot probes unblocking.

LEVEL FLIGHT

Set N1 as indicated in the table below, corresponding to MI = 0.75 (assumed temperature is ISA -10 ºC):

2. TCAS: TA ONLY .... Selected

Flight Level	Weight	N1	Pitch Attitude
FL 490	24,000 lb	97%	Between 1 and 4 degrees nose up
	22,000 lb	95%	
FL 450	30,000 lb	97%	
	22,000 lb	92%	
FL 410	38,000 lb	98%	
	30,000 lb	93%	
	22,000 lb	91%	
FL 370	38,000 lb	94%	
	30,000 lb	91%	
	22,000 lb	90%	
FL 330	38,000 lb	92%	
	30,000 lb	91%	
	22,000 lb	90%	
FL 310	38,000 lb	93%	
	30,000 lb	91%	
	22,000 lb	91%	

3. Advise ATC that both displayed altitude and XPDR-reported altitude may be unreliable and tightly monitor trajectory of closest airplanes.
4. When conditions permit, set N1 corresponding to cruise Mach = 0.75 at current flight altitude and airplane weight, using TAT as reference or standard atmosphere temperature if TAT is not usuable. (Performance Manual 5-05 pages 3 to 7).
5. Limit attitude to 4° nose up or less.

Continued on Next Page

CAUTION

VMO / MMO audio warning may be unreliable. If it is certain that the VMO / MMO audio warning is inappropriate, do not modify flight parameters.

If VMO / MMO audio warning sounds:

6. AUDIO WARN A / AUDIO WARN B circuit breakers............ PULLED

CAUTION

All aural warnings (STALL included) are inoperative except TCAS aural warning.

7. After a positive identification of the malfunction , continue the flight while complying with the following procedures for climb and descent phases:

CLIMB

1. N1 speed ....Climb power as per Performance Manual 4-50
2. Pitch attitude .... Between 4° and 5° nose up

If vertical speed drops below 100 ft / min:

3. Airplane..Level Off

DESCENT

Initiating the descent earlier than scheduled to recover non icing conditions is leFt to pilot's discretion.

CAUTION

If IAS goes down to 50 kt due to blocked pitot probes, expect loss of airspeed display on both EADI (if any):

- Do not apply PILOT AIR DATA COMPUTER INOPERATIVE procedure.
- Do not apply COPILOT AIR DATA COMPUTER INOPERATIVE procedure.
- Use ADI attitude and stand-by altimeter until pitot probes unblocking.

1. Start selector switches (all 3)...AIR START

ANTI-ICE:

2. ENG 1 and ENG 3 ANTI-ICE switches...ON
   - 30 seconds later:
     • ENG 2 ANTI-ICE switch..ON
   - 30 seconds later:
     • AIRFRAME ANTI-ICE switch ....NORMAL
3. N1 speed .... See table below

Continued on Next Page

TAT	Between -30°C and -20°C	Between -20°C and -10°C	Between -10°C and 0°C	0°C and above
N1	84%	81%	78%	73%

4. Airbrake handle.. Position 1
5. Pitch attitude .... Between 0º and 2º nose down
6. Vertical speed indicator ....Between -2,000 and -3,000 ft / min

NOTE

1 - Check airplane altitude frequently on the stand-by altimeter.

2 - If prior to the problems, flight was performed at a static temperature lower than the authorized minimum limit (see 1-15-9), descend as soon as possible until air-data indications are back to normal.

STATUS (2 blocked pitot probes):

INOPERATIVE / UNRELIABLE ITEMS	OPERATIVE / RELIABLE ITEMS
Basic fight parameters	
IAS / MI / TAS on both ADI and on stand-by instrument.	IRS Ground Speed in FMS CDU. GPS Ground Speed in FMS CDU.
	Pitch and roll attitude on both ADI.
Altitude reported by XPDR mode C.	Altitude displayed on both altimeters and stand-by instrument (max error +/- 600 ft). GPS altitude in NZ2000 FMS CDU (if any). VS on both VSI.
SAT, ISA deviation	TAT. Temperature data provided by AFIS / Operational flight plan / Weather briefing.
	Heading and Track.
	Wind data provided by AFIS / Operational flight plan / Weather briefing.
Warnings	
VMO / MMO audio warning	
Gear aural warning. Stall aural warning if AUDIO WARN C/Bs pulled	
Flight controls	
Automatic Slats extension.	Stall protection: IGN.
Roll Arthur position inconsistent with actual true flight condition.	Pitch Arthur (Arthur setting depends on the pressure (Pt) measured by the engine 1 probe).
Automatic flight control system	
AP, FD, and YD	
Mach Trim	
Engine	
	Engine primary parameters (N1, ITT, N2, FF) and controls.
Airplane Systems	
	All systems controls and displays

Continued on Next Page

NOTE

Descent should cause airspeed and total air temperature to increase thereby facilitating the pitot probes unblocking and a return to correct IAS indication after about 2 minutes.

NOTE

An indicated airspeed increasing in descent is a good evidence of the pitot probes unblocking.

7. After return to unblocked pitot probes situation, wait for 1 more minute then:

   - AP and YD..As required

8. AUDIO WARN A / AUDIO WARN B circuit breakers....... Re-engaged
9. TCAS...NORMAL - Checked
10. Start selector switches (all 3)...NORMAL
11. Airbrake handle...As required

ANTI-ICE:

- ANTI-ICE ENG 1 and 3 switches ....As required
- ANTI-ICE ENG 2 switch..As required
- ANTI-ICE AIRFRAME switch ....As required

6

FAILURE OF ALL THREE GENERATORS

GEN 1 AND GEN 2 GEN 3

Phase I

1. Bus C and D Switches.. OFF

NOTE

Autopilot will disengage.

2. Bus Tie Switch...FLIGHT NORMAL
3. Generator Switches..................... ATTEMPT RESETS (MAX 2 EACH)

Phase II

If all three generator lights remain illuminated:

Switch off as many systems as possible to reduce the battery load and maintain the current drain as low as possible:

4. BOOSTER Pumps Switches (All 3)
   Depending on the Fuel Envelope.. OFF
5. XFR Pumps Switches (All 3) .... OFF
6. WINDSHIELD Switches (All 3) .... OFF
7. ANTI-ICE:
   ENG and AIRFRAME Switches Depending
   on Flight Conditions... OFF
8. PITOT Switch (Pilot) Depending on Flight Conditions................. OFF
9. INTERIOR LIGHTS Switches .... OFF
10. EXTERIOR LIGHTS Switches .... OFF
11. Lighting Rheostat... FULLY CCW

7

LAND AS SOON AS POSSIBLE and avoid icing conditions.

CAUTION

The batteries in good condition will provide:

- 40 minutes of operation with an average load of 25 Amp per battery (approximate load after load shedding in non icing conditions).
- 20 minutes of operation with an average load of 50 Amp per battery (approximate load after load shedding in icing conditions).
- If conditions permit, the battery current is reduced by switching off:

  – All three BOOSTER pumps and three XFR pumps.
  – INS and Radio Navigation systems.
  – Windshield heat system.

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In icing conditons, it is imperative to keep the following switched on:

- Pilot windshield heating system.
- Engine and wing anti-ice systems

In icing conditions battery life is reduced by one minute for every minute that the systems are on.

NOTE

- The No. 1 Radio Navigation system is operable with battery power supply.
- The flap system, the rudder trim system and the emergency aileron trim system are inoperative (D bus switched off).
- The nose wheel steering system and the normal aileron trim system are inoperative (C bus switched off).
- Flap and nose wheel steering system loss.

AUTOPILOT NOSEDOWN HARDOVER

1. Autopilot ....DISENGAGE

HORIZONTAL TRIM STABILIZER RUNAWAY

WARNING

Audio warning: continuous clacker.

Warning panel: possible amber MISTRIM

Phase I:

1. Firmly hold the control column to maintain the desired path while actuating the normal pitch trim.
2. Actuate the TAILPLANE EMERG switch to trim the airplane.

FSI NOTE

This procedure is found in the Operating Manual, Book 1 of the Emergency Section (Not in AFM).

AILERON TRIM RUNAWAY

8 **NOTE**

With autopilot engaged, MISTRIM caution light is illuminated. Nevertheless, the flight-crew might not detect the failure before the trim reaches full travel as both yaw damper and autopilot counteract the roll.

1. Aileron trim.................................. Actuate in the opposite direction
2. Control wheel...Stop the roll
3. TRIM AILERON circuit breaker ....Pull
4. EMERG AILERON trim...Trim the airplane

FSI NOTE

This procedure is found in the Operating Manual, Book 1 of the Emergency Section (Not in AFM).

RUDDER TRIM RUNAWAY

NOTE

With autopilot engaged, the flight-crew might not detect the failure before the trim reaches full travel as both yaw damper and autopilot counteract the roll.

1. Rudder trim.................................... Actuate in the opposite direction
2. Control wheel.. Maintain wings level
3. Rudder pedals .... Cancel sideslip
4. Autopilot .... Disengage
5. Yaw damper.. OFF
6. Differential thrust...............................Reduce the effort on the pedals (as necessary)
7. Recommended maximum crosswind for landing..............23 KNOTS

FSI NOTE

This procedure is found in the Operating Manual, Book 1 of the Emergency Section (Not in AFM).

8

ENGINES INOPERATIVE

ALL ENGINES OUT CONDITION

Phase I

1. Bus C and D SWITCHES .... OFF
2. Communications.. VHF 1/ATC 1
3. Establish the aircraft within the In Flight Relight Envelope. (See page E-23)
4. Switch off as many services as necessary to decrease the battery load down to 50 amps per battery.
5. Relight all 3 engines per applicable procedure, Yellow Tab 3 or Yellow Tab 4.

Phase II

If no engine can be relighted:

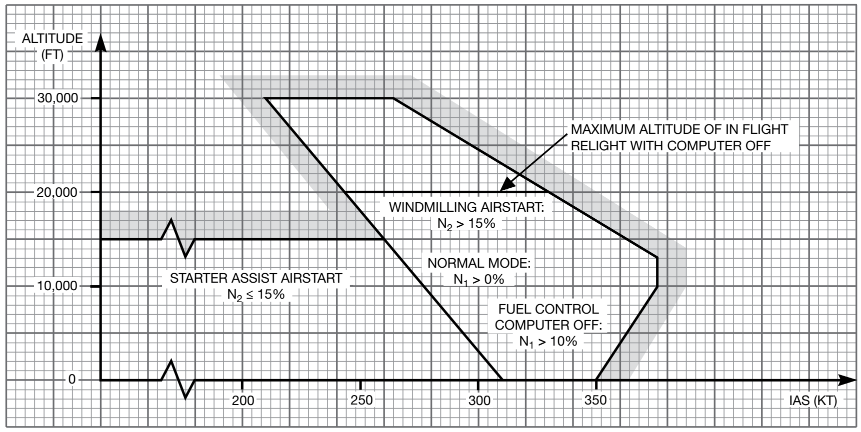
Be prepared to execute a Forced Landing or Ditching, Emergency section, Red Tab 11, page E-26 or E-27.

6. Standby Pump Switch ....ON
7. At VFE (200 KIAS) extend the slats using the emergency slat system if necessary.
8. If a forced landing is anticipated, extend the landing gear at VLO (190 KIAS), if possible, using the Landing Gear Free Fall procedure, Abnormal section; Yellow Tab 11, page A-22. This should take approximately two minutes.

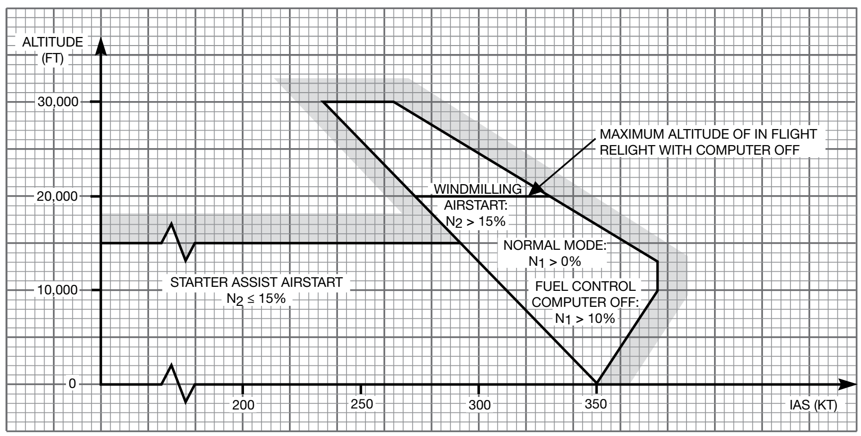
9

IN FLIGHT RELIGHT ENVELOPE

ENGINES 1 AND 3



ENGINE 2



9

APPROACH AND LANDING WITH TWO ENGINES INOPERATIVE

Preliminary Steps

Reduce the aircraft weight to minimum practical.

Determine the weight limitation for enroute climb gradient.

Determine the landing configuration and the landing distance/ field length additive.

a. Use S+20 Landing Distance Charts (Normal Procedures Checklist, Performance Tab, P-20, 21)

   1. Slats only .... VREF + 20 .... Add 1400/2300ft
   2. S +20 .... VREF +5 .... Add Charted Data
   3. Outboard stats only .... VREF + 25 .... Add 1580/2600 FT
   4. Outboard S+20 .... VREF +10 .... Add 180/300 FT

b. If brakes are supplied by the #2 hydraulic system, increase landing distance/landing field length by 30%.

1. Fuel System Situation .... CHECKED
2. Bus Tie Switch .... TIED
3. Limit Generator Load .... 300 AMPS MAX
4. Crew and Cabin Bleed Air Switches .... OFF
5. With #1 and #2 Engines Inoperative .... Brakes #2 OFF
6. Hydraulic Standby Pump Switch .... ON
7. Avoid icing conditions.

Approach

With engines #1 and #2 inoperative, refer to checklist (A).

10 With engines #1 and #3 inoperative, or engines #2 and #3 inoperative, refer to checklist (B).

(A) Engines #1 and #2 inoperative.

1. Slats (200 KIAS) .... Emergency Slats Switch

   When committed for landing:

2. Landing Gear (190 KIAS) .... EXTEND (YELLOW TAB 11)
   a. Normal Gear Handle .... DOWN
   b. EMERG GEAR Handle .... PULL
   c. Main Gear Manual Release Handles .... PULL
   d. Nose Gear Manual Release Handle.................................... PULL
   e. Brake Selector .... Brakes #2 OFF
3. Slat Flap Handle .... S+20, or as previously determined
4. Approach Speed:

   Outboard slats only .... VREF + 25 KNOTS

   Outboard S+20 .... VREF + 10 KNOTS

Use moderate braking with brakes supplied by the #2 hydraulic system.

Continued on Next Page

(B) Engines #1 and #3 inoperative, or engines #2 and #3 inoperative.

1. Slats..EXTENDED
2. At no less than 1000 FT AGL, decision ....GO AROUND OR LAND

   When committed for landing:
3. Landing Gear .... DOWN
4. Slat Flap Handle .... S + Flaps 20, or as previously determined
5. Approach Speed:
   Slats Only...VREF + 20 KNOTS
   S+20 ....VREF + 5 KNOTS

After Touchdown

Normal deceleration procedure if braking with antiskid is available and if thrust reverser is available.

GO-AROUND WITH TWO ENGINES INOPERATIVE

CAUTION

The decision to land or go-around must be made at or above 1,000 feet AGL.

The altitude loss associated with the go-around procedure is approximately 500 feet.

The landing gear cannot be retracted with #1 and #2 engines inoperative.

1. Take-off Thrust...SET
2. Landing Gear Handle.. UP

Accelerate while in descent on normal slope.

At VREF + 25 knots

3. Slat Flap Handle .... CLEAN

Accelerate to the enroute climb speed, and initiate climb.

EMERGENCY CONDITIONS

FORCED LANDING

Preliminary Steps:

1. Transmission of Distress Signal .... “MAYDAY”
2. Transponder Code 7700
3. Passenger Briefing and Preparation .... COMPLETE
4. Fasten Seat Belt/No Smoking Sign .... ON
5. Cockpit Jump Seat (If Possible) .... STOWED

Approach:

6. Airframe Anti-ice .... OFF
7. Crew Bleed Air Switch .... OFF
8. Cabin Bleed Air Switch .... OFF
9. HP Bleed Switches .... OFF
10. PRV Switch .... OFF
11. Cabin Pressure Switch .... DUMP
12. Manual Pressurization Knob .... UP
13. Landing Gear .... EXTENDED
14. Slats/Flaps .... SLATS + FLAPS 48
15. Approach Speed .... VREF

Just Before Touchdown:

16. Vertical Speed .... APPROX 300 FT/MIN
17. Generator Switches .... OFF
18. Battery Switches .... OFF
19. Power Levers .... CUTOFF

After the aircraft has come to rest:

20. Fire Handles (All Three) .... PULL

11

21. Engine Fire Extinguisher (All Three) .... POSITION 2
22. Cockpit Jump Seat .... STOWED/AISLE CLEARED

Use emergency exits and cabin access door to evacuate the aircraft.

DITCHING

Preliminary Steps:

1. Transmission of Distress Signal..“MAYDAY”
2. Transponder Code... 7700
3. Passenger Briefing and Preparation................................COMPLETE
4. Life Jackets..DONNED/CHECKED
5. Fasten Seat Belt/No Smoking Sign ....ON
6. Cockpit Jump Seat (If Possible) ....STOWED
7. Audio Warning Circuit Breakers... PULLED

Approach (Parallel to Main Swell):

8. Airframe Anti-ice Switch .... OFF
9. Crew Bleed Air Switch .... OFF
10. Cabin Bleed Air Switch .... OFF
11. Bag Air Switch .... OFF
12. ECU Inlet Control Door Handle... PULL
13. HP Bleeds .... OFF
14. PRV Switch .... OFF
15. Cabin Pressurization Selector Switch ....DUMP
16. Manual Pressurization Knob.. UP
17. Landing Gear .... UP
18. Slats/Flaps .... SLATS + FLAPS 48
19. Airspeed.. VREF

Prior to contact:

20. Vertical Speed..APPROX 300 FT/MIN
21. Generator Switches .... OFF
22. Battery Switches.. OFF

Ditch the aircraft on the crest and parallel to swell at the slowest practical speed and with a noseup attitude of 11 to 13°.

After touchdown:

23. Power Levers ....CUTOFF
24. Engine Fire Handles (All Three).. PULL 11
25. Cockpit Jump Seat..................................... STOWED/AISLE CLEAR

CAUTION

Do not open the main cabin door.

Use emergency exits to evacuate the aircraft.

BOMB ON BOARD

To avoid the activation of an altitude sensitive or timer ignition bomb, the following procedure focuses on:

- Making the cabin altitude not exceed the value at which the bomb was discovered
- Trying to minimize the flight time.

To reduce the effects of explosions by helping the blast to go outwards, the flight-crew should maintain approximately 1 psi differential pressure (which corresponds to a 2,500 ft difference between the airplane altitude and the cabin altitude).

Procedure

1. Airplane..LEVEL OFF
2. FASTEN BELTS and no smoking light pushbuttons.................... ON

   Seat belts must be fastened, properly adjusted and sufficiently tightened. The crew must also put on their harnesses and check that they are locked.
3. Transmission of distress signal... MAYDAY
4. ATC transponder..MAYDAY CODE
5. UP–DN control .... Between 1 and 2 o'clock minimum 30 seconds
6. Cabin pressure selector switch .... MAN
7. Cabin vertical speed..MAINTAIN TO 0
8. Divert to the nearest suitable airport in the shortest time.
9. Descent to altitude = cabin altitude + 2,500 ft (Δp = 1 psi) or safety altitude.
10. When at cabin altitude + 2,500 ft:

    Continue descent maintaining Δp CABIN = 1 psi.

CAUTION

If structural integrity is questioned:

- Limit airspeed value to lowest practical value,
- Avoid high manuevering load factors

12

Continued on Next Page

NOTE

Approach and landing are performed using the normal procedures and checklist.

Before Landing

10. Cabin pressure selector switch ....AUTO
11. UP–DN control .... DN POSITION (Full CCW)

After landing and when airplane stopped in a remote area:

12. EMERGENCY EVACUATION procedure............... ACCOMPLISHED

FSI NOTE

This procedure is found in the Operating Manual, Book 1 of the Emergency Section (Not in AFM).

12

VOLCANIC ASH ENCOUNTER

WARNING

EXIT ASH CLOUD AS QUICKLY AS POSSIBLE. DO NOT ATTEMPT TO CLIMB OUT OF THE ASH CLOUD.

CAUTION

Weather radar does not detect volcanic ash.

1. Land at the nearest suitable airport.
2. Course ....Reverse, 180° turn (1)
3. ATC transponder.. 7700
4. MAYDAY message...TRANSMIT
5. Crew oxygen masks ....DONNED - 100 % + EMERGENCY
6. Smoke goggles................................DONNED - VENT VALVE OPEN
7. Start selector switches (all 3)... AIR START
   - IGN lights ....ON
8. ANTI-ICE ENG switches (all 3)...ON (2)
9. Power levers: one by one ....SMOOTHLY REDUCE THRUST TO LOWEST ITT (3)

If unable to maintain altitude:

- Fly a drift down descent.

10. Power levers ....AVOID RAPID THRUST CHANGES (3)
11. Engines parameters.. MONITOR

If engine surge is evidenced:

- Corresponding engine ....SMOOTHLY REDUCE THRUST UNTIL ENGINE STABILIZES
- Be prepared to apply:
  - ENGINE FAILURE IN FLIGHT procedure Yellow tab Tab 2, A-2
  or
  - ALL ENGINES OUT CONDITION procedure Red tab Tab 9, E-22 (4).

13

Continued on Next Page

12. Airspeed indications .... MONITOR

If airspeed indications become unreliable:

- Apply UNRELIABLE AIRSPEEDS AT HIGH ALTITUDE procedure Red tab Tab 6, E-13.

If smoke, acrid odor or volcanic dust fill the cabin:

- NO SMOKING light pushbutton..ON
- PASSENGER OXYGEN controllerOVERRIDE (5)
- Passenger masks...DONNED - CHECKED

Technical Status

Many airplane systems can be severely affected.

Operational Status

The pilot should land at the nearest suitable airport.

Technical Explanations

For additional explanations, refer to “operations in contaminated airspace” procedure (section 2-80).

FSI NOTE

This procedure is found in the Operating Manual, Book 1 of the Emergency Section (Not in AFM).

Expanded Explanations

(1) 180 turn:

Considering that an ash cloud can extend for hundred of miles, a 180 turn in descent is the best escape strategy. Depending on the situation, an emergency descent may be necessary.

(2) ANTI ICE: ENG switches (all 3):

This improves the engine stall margins by increasing the bleed air flow, and increases the fuel /air ratio in the combustion chamber.

(3) Thrust levers:

Keeping ITT at the lowest value lowers the fused ash buildup on turbine blades and hot section components.
Rapid thrust variation must be avoided since engine surge margins may be reduced.

(4) Engine relight:

If engine relight is attempted, it could take longer than normal to reach the idle thrust due to the combined effects of high altitude and volcanic ash ingestion.

(5) PASSENGER OXYGEN controller:

Select OVERRIDE to ensure the masks boxes to open and oxygen to be supplied to the passenger masks

STALL RECOVERY

Phase I

SIMULTANEOUSLY and SMOOTHLY:

1. Autopilot ....DISENGAGE
2. Pitch..NOSE DOWN
3. Bank...WINGS LEVEL
4. Power levers .... FULL FORWARD
5. Airbrakes.. POSITION 0
6. Pitch trim ....AS REQUIRED

Phase II

7. Pitch attitude .... CLOSE TO HORIZON
8. Thrust...ADJUST
9. Speed and FD modes..AS REQUIRED
10. Autopilot .... AS REQUIRED

FSI NOTE

This procedure is found in the Operating Manual, Book 1 of the Emergency Section (Not in AFM).

UNUSUAL ATTITUDE RECOVERY - NOSE UP

Phase I

1. Autopilot ....DISENGAGE
2. Power levers ....FULL FORWARD
3. Bank the aircraft by the shortest way....................80° TO 90° BANK
4. Airbrakes.. POSITION 0

When aircraft nose is close to horizon:

5. Bank..WINGS LEVEL
6. Power levers ....ADJUST

Phase II

7. Speed and FD modes...AS REQUIRED
8. Autopilot .... AS REQUIRED

FSI NOTE

This procedure is found in the Operating Manual, Book 1 of the Emergency Section (Not in AFM).

UNUSUAL ATTITUDE RECOVERY - NOSE DOWN

Phase I

1. Autopilot ....DISENGAGE
2. Power levers..Idle
3. Bank...Wings level
4. Airbrakes...AS REQUIRED
5. Nose up pitch ....SMOOTHLY APPLY

When aircraft nose is close to horizon:

6. Airbrakes.. POSITION 0
7. Power levers ....ADJUST

Phase II

8. Speed and FD modes...AS REQUIRED
9. Autopilot ....AS REQUIRED

FSI NOTE

This procedure is found in the Operating Manual, Book 1 of the Emergency Section (Not in AFM).

PILOT INCAPACITATION

If a pilot becomes incapacitated, the remaining pilot must:

1. Maintain control of the aircraft by:
   - Taking over the controls,
   - Selecting the AP XFR switch button on the valid pilot side,
   - Engaging the AP as soon as possible.
     If transfer of the autopilot control is necessary, verify control modes. The autopilot will remain engaged, however, the modes may possibly revert to basic modes or not be the same as the original side prior to the transfer.
2. Check position of essential controls and switches.
3. Inform ATC and declare an emergency.
   Communications are preferably performed using the headset.
4. Request the help of a cabin crew or a passenger to assist the incapacitated pilot:
   - Pull the pilot back by the shoulders,
   - Tighten and manually lock the shoulder harness of the incapacitated pilot,
   - Place and secure the hands under the belt harness,
   - Move the seat completely aft,
   - Recline the seat back if possible,
   - Lift each knee to remove the pilot's feet from the rudder pedals,
   - Move the rudder pedals of the incapacitated pilot fully forward,
   - Perform a medical assessment of the incapacitated pilot (including vital signs),
   - Administer first aid as necessary,
   - Consider use of the oxygen mask or therapeutic oxygen and mask,
   - Contact medical assistance if time and environmental conditions permit,
   - Remove the incapacitated pilot from the seat if phase of flight and environmental conditions permit.

Continued on Next Page

5. Reorganize the cockpit workload:
   - Distribute the workload among the remaining crew if possible,
   - Perform checklists earlier than normal,
   - Request radar vectoring whenever possible,
   - Achieve landing configuration earlier than normal.
6. Divert to the nearest suitable airport depending on medical emergency level and flight conditions as required.

NOTE

It takes two people to remove the weight of an unconscious person from a seat without endangering any controls and switches.

If it is not possible to remove the incapacitated pilot, the cabin crew or any medically qualified passenger must remain in the cockpit to take care of and observe the incapacitated pilot.

FSI NOTE

This procedure is found in the Operating Manual, Book 1 of the Emergency Section (Not in AFM).

EMERGENCY EVACUATION

Phase I

1. Park Brake .... SET
2. ATC—Distress Transmission .... NOTIFY
3. Passengers .... INSTRUCTED

Phase II

4. Fire Pull Handles (All 3) .... PULL
5. GEN and BAT Switches (All 5) .... OFF
6. EMERG LIGHTS Switch .... ON
7. Engine Fire Extinguisher DISCH Switch (All 3) .... POSITION 2
8. APU Fire Extinguisher DISCH Switch .... POSITION 1
9. Third Cockpit Seat .... STOWED
10. Evacuation .... INITIATE

FSI NOTE

This procedure is found in the Operating Manual, Book 1 of the Emergency Section (Not in AFM).

ABNORMAL PROCEDURES

Procedures in this section address foreseeable situations involving failures, in which the system's redundancy or selection of an alternate system will maintain an acceptable level of air worthiness.

I. IMMEDIATE ACTION—There are no memory items involved with these procedures. When a failure occurs, the crew should assess its priority in relationship to the immediate effect on aircraft controllability and the continuance of the planned flight path.

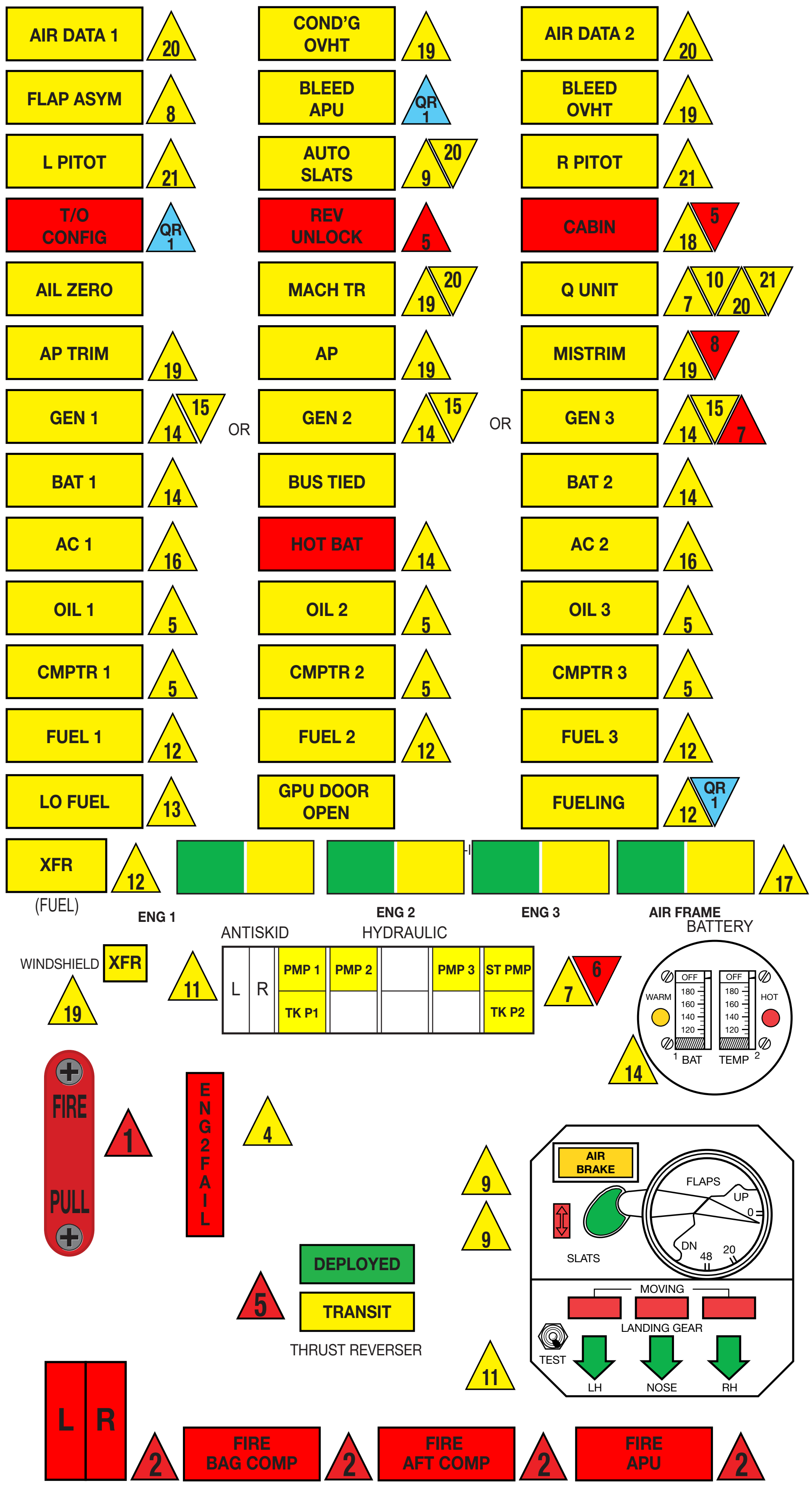
   Procedures that could have an immediate effect, but involve actions that are fundamental to basic airmanship, are:

   - Air Data System Failures—Refer to standby instruments or transfer control.
   - Engine Vibration (Actual)—Reduce thrust.
   - Pitch Trim Runaway—Override or disconnect.

II. SPECIFIC PRIORITIES—Where there is no immediate action required, it is advisable to complete a Normal Checklist if in progress or due prior to calling for the Abnormal Checklist.

   Because of the interrelationship of systems, a failure in one can have an effect on other systems.

   The system that can produce multiple annunciations are (1) Engine, (2) Electric, (3) Hydraulic and should be dealt with in priority as numbered.



AIR DATA 1
20
COND'G OVHT
19
AIR DATA 2
20
FLAP ASYM
8
BLEED APU
QR 1
BLEED OVHT
19
L PITOT
21
AUTO SLATS
9
20
R PITOT
21
T/O CONFIG
QR 1
REV UNLOCK
5
CABIN
18
5
AIL ZERO
MACH TR
19
20
Q UNIT
7
10
20
21
AP TRIM
19
AP
19
MISTRIM
19
8
GEN 1
14
15
OR
GEN 2
14
15
OR
GEN 3
14
15
7
BAT 1
14
BUS TIED
BAT 2
14
AC 1
16
HOT BAT
14
AC 2
16
OIL 1
5
OIL 2
5
OIL 3
5
CMPTR 1
5
CMPTR 2
5
CMPTR 3
5
FUEL 1
12
FUEL 2
12
FUEL 3
12
LO FUEL
13
GPU DOOR OPEN
FUELING
12
QR 1
XFR
12
(FUEL)
ENG 1
ENG 2
ENG 3
AIR FRAME
17
BATTERY
ANTISKID
HYDRAULIC
WINDSHIELD
XFR
19
11
L
R
PMP 1
PMP 2
PMP 3
ST PMP
TK P1
TK P2
7
6
OFF
180
160
140
120
WARM
HOT
1 BAT
TEMP 2
14
FIRE
PULL
1
ENG 2 FAIL
4
9
9
AIR BRAKE
FLAPS
UP
0
DN
48
20
SLATS
DEPLOYED
TRANSIT
THRUST REVERSER
5
MOVING
LANDING GEAR
TEST
LH
NOSE
RH
11
L
R
2
FIRE BAG COMP
2
FIRE AFT COMP
2
FIRE APU
2

ABNORMAL PROCEDURES

ENGINE	ENGINE FAILURE BEFORE V1 ENGINE FAILURE AFTER V1	1
	INFLIGHT ENGINE FAILURE INFLIGHT RELIGHT ENVELOPE	2
	AIRSTART	3
	FLAMEOUT AND HIGH SPEED AIRSTART ENGINE NO. 2 AIR INLET DOOR OPEN	4
	ENGINE COMPUTER INOPERATIVE ENGINE OIL	5
ONE ENGINE INOPERATIVE PROCEDURE	ONE ENGINE INOPERATIVE APPROACH/LANDING/GO-AROUND (APPROACH CONFIGURATION SLATS OR S+20) APPROACH LANDING	6
HYDRAULICS	UNWANTED OPERATION OF STANDBY PUMP LOSS OF NO. 1 HYDRAULIC SYSTEM DEPRESSURIZATION OF HYDRAULIC RESERVOIRS LOSS OF NO. 2 HYDRAULIC SYSTEM LOSS OF NO. 3 ENGINE DRIVEN PUMP	7
FLIGHT CONTROLS	FLAP SYSTEM JAMMING OR A SYMMETRY AILERON SYSTEM JAMMING RUDDER JAMMING	8
	AIRBRAKES DO NOT RETRACT SLAT MONITORING SYSTEM MALFUNCTION SLAT SYSTEM MALFUNCTION	9
	Q UNIT INOPERATIVE ELEVATOR LANDING INOPERATIVE STABILIZER LANDING	10
LANDING GEAR STEERING BRAKES	LANDING GEAR RETRACTION MALFUNCTION LANDING GEAR EXTENSION MALFUNCTION NOSEWHEEL STEERING FAILURE NOSEWHEEL SHIMMY NO. 1 BRAKE SYSTEM OR ANTISKID INOPERATIVE NO. 1 AND NO. 2 BRAKE SYSTEMS INOPERATIVE	11

FUEL	LOW BOOSTER PUMP PRESSURE	
	FUEL TRANSFER SYSTEM INOPERATIVE	**12**
	FUELING LIGHT ON INFLIGHT	
	WING TANK LEVEL ABNORMALLY LOW	
	FUEL FEEDER TANK LEVEL LOW	**13**
	FEEDER TANK LEVEL HIGH	
ELECTRICAL	TWO GENERATORS INOPERATIVE	**14**
	BATTERY FAILURE	
	BATTERY OVERHEAT	
	ONE GENERATORS INOPERATIVE	**15**
	INVERTER FAILURE	**16**
	FAILURE OF BOTH INVERTERS	
ANTI-ICE	ENGINE ANTI-ICE SYSTEM INOPERATIVE	**17**
	ENGINE ANTI-ICE SYSTEM UNWANTED OPERATION	
	AIRFRAME ANTI-ICE SYSTEM INOPERATIVE	
	AIRFRAME ANTI-ICE SYSTEM UNWANTED OPERATION	
	ICE PROTECTION—LATE ACTIVATION	
PRESSURIZATION / AIR CONDITIONING	HIGH CABIN ALTITUDE OR SLOW DEPRESSURIZATION	**18**
	HIGH CABIN PRESSURE DIFFERENTIAL	
	IMPROPER CABIN VERTICAL SPEED	
	NO AUTOMATIC PRESENTATION OF PASSENGER MASKS	
	CABIN DOOR UNLOCKED	
	CABIN AIR CONDITIONING OVERHEAT	**19**
	BLEED AIR OVERHEAT	
WINDSHIELD	CRACKED WINDSHIELD PANE	
	WINDSHIELD HEAT SYSTEM MALFUNCTION	
AUTOPILOT/ TRIM	AUTOPILOT FAILURE	
	AUTOPILOT PITCH TRIM INOPERATIVE	
	OUT OF TRIM CONDITION	
	MACH TRIM INOPERATIVE	
FLIGHT INSTRUMENTS	PILOT AIR DATA COMPUTER INOPERATIVE	**20**

PITOT STATIC	PILOT PITOT-STATIC SYSTEM MALFUNCTION COPILOT PITOT-STATIC SYSTEM MALFUNCTION FLIGHT WITH SUSPECTED BLOCKED PITOT PROBES	**21**
EFIS	EFIS MALFUNCTIONS	**22**
	EADI DISPLAY FAILURE EHSI DISPLAY FAILURE SIMULTANEOUS FAILURE OF EADI AND EHSI DISPLAYS ON SAME SIDE SUCCESSIVE FAILURE OF EADI AND EHSI DISPLAYS ON SAME SIDE FAILURE MONITOR FLAGS (ALL RED)	**23**
COLLINS EFIS	COMPARISON FLAGS (ALL AMBER) ATTITUDE FAIL HEADING FAIL MPU FAILURE	**24**
	DCP FAILURE RADIO ALTIMETER FAILURE FLIGHT DIRECTOR FAILURE ATTITUDE COMPARISON MONITOR HEADING COMPARISON MONITOR	**25**
	IAS COMPARISON MONITOR LOC OR GS COMPARISON MONITOR RA COMPARISON MONITOR	**26**

ENGINE

ENGINE FAILURE BEFORE V1

Abort the takeoff:

1. Brakes....................MAXIMUM
2. Power Levers.................... IDLE
3. Airbrake Handle....................POSITION 2
4. Thrust Reverser.................... DEPLOYED

ENGINE FAILURE AFTER V1

Continue the takeoff:

1. At VR....................ROTATE NORMALLY
2. Take Off Attitude....................SET
3. Airspeed.................... MAINTAIN V2
4. Positive Rate of Climb....................GEAR UP

CAUTION

If the engine failure occurs at a speed above V2, maintain that speed.

Above 400 feet AGL:

5. Level Flight Acceleration.................... INITIATE
6. Airspeed:
   a. Slats + 20°....................V2 + 15 KNOTS
   b. Slats Only....................V2 + 25 KNOTS
7. Slat/Flap Handle.................... CLEAN
8. Climb Speed....................MAINTAIN 1.5 VS

5 minutes after brake release:

9. Maximum Continuous Thrust....................SET
10. After Takeoff Checklist....................COMPLETE
11. Refer to Inflight Engine Failure checklist, Yellow Tab 2, page A-2.

INFLIGHT ENGINE FAILURE

Identify the failed engine:

OR

If engine integrity is questionable, or if airstart is unsuccessful:

- Complete engine shutdown as follows:

If engine integrity is not questionable:

- Proceed to Airstart Checklist, Tab 3, page A-4.

1. Power Lever........IDLE FOR 1 MINUTE IF POSSIBLE,THENCUTOFF
2. Booster Pump.. OFF
3. Generator Switch...OFF*

NOTE

In icing conditions, if the #1 or #2 engine has been shutdown, continue to operate respective engine anti-ice system.

*4. Engine Anti-ice Switch.. OFF
5. HP Bleed Switch.. OFF
6. PRV Switch (With #2 Engine Inoperative)................................... OFF
7. Fire Handle... PULL

If engine #3 is shutdown:

8. Bus Tie Switch.. TIED
9. Hydraulic Standby Pump Switch........................ON (AS REQUIRED)

OR

One Engine Inoperative Approach/Landing/Go-Around Procedures:

- Yellow Tab 6, page A-12

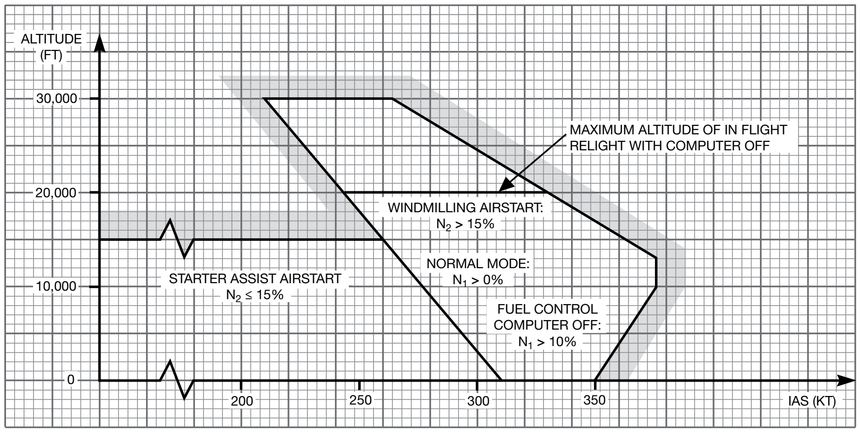
Approach And Landing With Two Engines Inoperative:

- Red Tab 10, pages E-24 and E-25

INFLIGHT RELIGHT ENVELOPE

2

ENGINES 1 AND 3



ENGINE 2

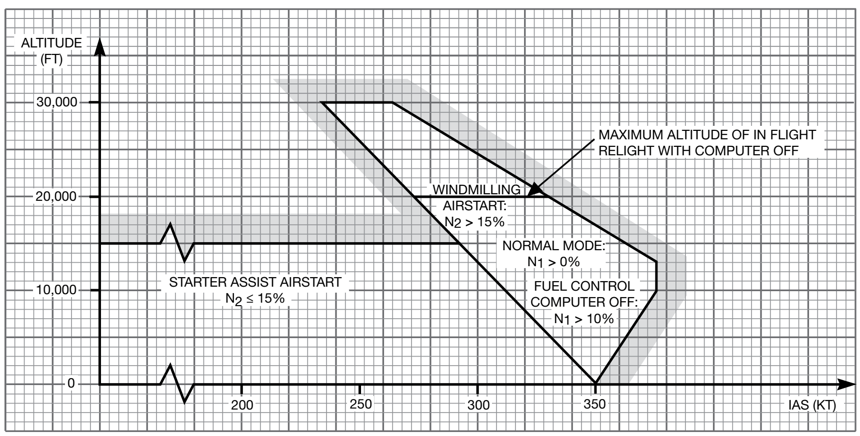


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FIG A-1
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AIRSTART

3

WARNING

DO NOT ATTEMPT TO RELIGHT AN ENGINE AFTER AN ENGINE FIRE OR IF THE ENGINE INTEGRITY IS QUESTIONABLE.

CAUTION

Wait 10 seconds between two consecutive airstart attempts. Do not make more than three successive airstart attempts.

Preliminary Steps:

*Establish aircraft within the Airstart envelope; Tab 2, page A-3.

1. Fire Handle....................IN
2. Power Lever....................CUTOFF
3. Generator Switch....................ON
4. Engine Computer Switch....................ON
5. Booster Pump....................ON
6. Engine Anti-ice Switch....................OFF (IF POSSIBLE)
7. Airframe Anti-ice Switch....................OFF (IF POSSIBLE)
8. Bus Tie Switch....................TIED

Determine airstart conditions:

OR

If N2 speed greater than 15% and indication of N1 rotation:

- Use procedures A, Windmilling Airstart, page A-5.

If N2 speed not greater than 15%:

- Use procedure B, Starter Assist Airstart, page A-6.

If Engine Computer is inoperative:

- Use procedure C, Airstart with Fuel Control Computer Off, page A-7.

NOTE

Abort the airstart if:

- Oil pressure does not rise within 10 seconds after light-off.
- ITT does not rise within 10 seconds after light-off.
- ITT is rising rapidly and approaching 907°C limit.
- N1 remains close to zero when N2 = 20%.
- N2 speed is not rising rapidly and smoothly after light-off.
- During an Airstart with Fuel Control Computer Off, the N1 exceeds 80% with the power lever at idle.

Continued on Next Page

NOTE

If ignition light remains on through N2 speed greater than 50%:

1. Start Selector Switch ....MOTOR START STOP

   Check that the IGN light is out.

2. Start Selector .... GROUND START

(A) Windmilling Airstart

1. Start Selector Switch ....AIRSTART (IGN LIGHT ON)
2. Power Lever.. IDLE

   Check that the ITT rises within 10 seconds.

When N2 speed greater than 50%:

3. Start Selector Switch..................GROUND START (IGN LIGHT OUT)

   Check that the IGNITION, GENERATOR, and OIL lights are out.

4. Engine Instruments..CHECKED
5. Bus Tie Switch..FLIGHT NORMAL

   Check that the BUS TIED light is out.

NOTE

To abort airstart:

1. Power Lever ....CUT-OFF
2. Start Selector Switch .... MOTOR START STOP
3. Return to Inflight Engine Failure checklist, Yellow tab 2 page A-2.

Continued on Next Page

AIRSTART (Cont)

(B) Start Assist Airstart

NOTE

A starter assist airstart may cause disengagement of the autopilot system.

1. Start Selector Switch..............................AIR START (IGN LIGHT ON)
2. Start Pushbutton................ DEPRESSED (LESS THAN 2 SECONDS)

When N2 speed is 12-15% and indication of N1 rotation:

3. Power Lever.. IDLE

   Check that the ITT rises within 10 seconds, and that N1, fuel flow, and oil pressure are rising.

When N2 speed is greater than 50%:

4. Start Selector Switch..................GROUND START (IGN LIGHT OUT)

   When the N2 stabilizes, check that the generator, oil, and the pump lights are out.
5. Engine Instruments...CHECKED
6. Bus Tie Switch...FLIGHT NORMAL

   Check that the BUS TIED light is out.

NOTE

To abort airstart:

1. Power Lever .... CUT-OFF
2. Start Selector Switch ....MOTOR START STOP
3. Return to Inflight Engine Failure checklist, Yellow Tab 2, page A-2.

Continued on Next Page

Ⓒ Airstart with Fuel Control Computer Off

Windmilling Airstart

Use Windmilling Airstart procedures, page A-5, for normal mode, but with N2 speed greater than 15% and N1 speed greater than 10%.

Starter Assist Airstart

1. Start Selector Switch..............................AIR START (IGN LIGHT ON) 3
2. Start Pushbutton..................... DEPRESS (LESS THAN 2 SECONDS)

When N2 speed is 10% and N1 rotation is observed:

3. Power Lever.. IDLE

   Check that ITT rises within 10 seconds.

When N2 speed is 50%:

4. Start Selector Switch.......................................MOTOR START STOP

   Check that the ignition, generator, and the oil lights are out.

5. Engine Instruments..CHECKED
6. Start Selector Switch..GROUND START
7. Bus Tie Switch..FLIGHT NORMAL

   Check that the BUS TIED light is out.

NOTE

To abort airstart:

1. Power Lever ....CUT-OFF
2. Start Selector Switch.............MOTOR START STOP
3. Return to Inflight Engine Failure checklist, Yellow Tab 2, page A-2.

FLAME-OUT AND HIGH SPEED AIRSTART

WARNING

DO NOT ATTEMPT TO RELIGHT AN ENGINE AFTER AN ENGINE FIRE OR IF THE ENGINE INTEGRITY IS
4 QUESTIONABLE.

CAUTION

Wait 10 seconds between two consecutive airstart attempts. Do not make more than three successive airstart attempts.

N2 RPM 15% or above

NOTE

This immediate airstart may be attempted at high altitude even at altitudes above the maximum start envelope.

1. Power Lever....................................IMMEDIATELY TO IDLE
2. Start Selector Switch..........................AIR START (IGN LIGHT ON)

Check that the ITT rises within 10 seconds.

3. Power Lever....................................ADVANCE

After relight:

4. Start Selector Switch..........................GROUND START (IGN LIGHT OUT)
5. Engine Instruments.............................CHECKED

ENGINE NO. 2 AIR INLET DOOR OPEN

- If the failure occurs on the ground before V1, abort the takeoff.
- If the failure occurs on the ground after V1, continue the takeoff.

In Flight:

1. Engine #2 Power Lever... IDLE

   If the engine surges or abnormal conditions are observed, shutdown the #2 engine.

Engine Shutdown Procedure:

1. Power Lever.. CUTOFF
2. Booster Pump... OFF
3. Generator Switch.. OFF

***NOTE**

In icing conditions, continue to operate #2 engine anti-ice system.

*4. Engine Anti-ice Switch.. OFF
5. HP Bleed Switch... OFF
6. PRV Switch (With #2 Engine Inoperative)................................... OFF
7. Fire Handle.. PULL

OR

One Engine Inoperative Approach/Landing/Go-Around Procedures:

• Yellow Tab 6, page A-12

Approach And Landing With Two Engines Inoperative:

• Red Tab 10, pages E-24 and E-25

ENGINE COMPUTER INOPERATIVE

CMPTR

NOTE

Prior to step No. 1, if operationally feasible, reduce power on affected engine below 80% N1.

1. Engine Computer Switch..OFF THEN ON

If the COMPTR light remains on:

2. Do not let the ITT of affected engine exceed indicated ITT of other engines.
5 3. Avoid rapid displacements of power lever.

CAUTION

- Maximum thrust may not be obtained.
- Idle thrust may be higher than normal.
- At a given N1 speed, fuel flow is approximately 5% higher.
- Surge Bleed valve will assume the 1/3 open position.

ENGINE OIL

OIL

NOTE

The illumination of the OIL light indicates a low oil pressure condition, or the presence of metal chips in the lubrication system.

If the oil pressure is normal:

1. Monitor Oil Pressure and Temperature.
2. Reduce Power if Possible.

If the oil pressure is less than 25 PSI:

1. Reduce power.
2. Shutdown the engine as soon as possible.

Engine Shutdown Procedure: 5

1. Power Lever..CUTOFF
2. Booster Pump..OFF
3. Generator Switch..OFF

*NOTE

In icing conditions, if the #1 or #2 engine has been shutdown, continue to operate respective engine anti-ice system.

*4. Engine Anti-ice Switch..OFF
5. HP Bleed Switch..OFF
6. PRV Switch (With #2 Engine Inoperative)..OFF
7. Fire Handle..PULL

If engine #3 is shutdown:

8. Bus Tie Switch..TIED
9. Hydraulic Standby Pump Switch........................ON (AS REQUIRED)

OR

One Engine Inoperative Approach/Landing/Go-Around Procedures:

• Yellow Tab 6, page A-12

Approach And Landing With Two Engines Inoperative:

• Red Tab 10, pages E-24 and E-25

ONE ENGINE INOPERATIVE PROCEDURE

ONE ENGINE INOPERATIVE APPROACH/ LANDING/GO-AROUND (APPROACH CONFIGURATION SLATS OR S+20)

1. Hydraulic Implications ....CHECK
2. Electrical Implications..CHECK
3. Landing Distance and Climb Requirements ....CHECK

Approach

4. Slat/Flap Handle .... Slats or S + 20
5. Landing Gear ....DOWN
6. Approach Speed..SET

   S+20 ....VREF + 5 KNOTS

   Slats Only...VREF + 20 KNOTS
7. GPWS (Landing less than S+48) ....FLAP OVRD
8. Approach and Landing Checklists.................................COMPLETE

NOTE

When landing is assured and the possibility of a go-around is excluded, the slats / flaps may be extended to:

Approach with S+ 20 ....S + 48 (VREF)

Approach with Slats ....S + 20 (VREF+5)

Go-Around

1. Power Levers...SET T/O N1
2. Pitch Attitude...14°
3. Landing Gear .... UP
4. Climb Speed..SET

   S + 20 ....VREF + 5 KNOTS

   Slats Only...VREF + 20 KNOTS

At no lower than 400 feet AGL:

5. Level Flight Acceleration.. INITIATE

   At VREF + 15 Knots... SLATS ONLY

   At VREF + 25 Knots.. CLEAN
6. Enroute Climb Speed...Attain 1.5 V5
7. After T/O Checklist ....COMPLETE

Continued on Next Page

APPROACH

1. Slats/Flaps .... SET
2. Seat Belt Sign/No Smoking Signs .... ON
3. Anti-Icing .... AS REQUIRED
4. Radios .... SET FOR APPROACH
5. Passenger Briefing .... COMPLETED
6. Start Selector Switches .... AIRSTART
7. Ignition Lights .... (3) ON

LANDING

1. Landing Gear .... DOWN/THREE GREEN
2. Brake Selector .... #1 ON
3. Antiskid .... TESTED
4. Hydraulics (Pressure/Quantity) .... CHECKED
5. Test/Stall (Aircraft without SB166) .... STALL 1/STALL 2 TESTED 6
   Test/Stall (Aircraft with SB166) .... AUTO SLAT LIGHT OUT
6. Flaps .... SET
7. No Smoking Sign .... ON
8. Taxi Light .... ON
9. Windshield Wipers .... AS REQUIRED
10. Airbrakes .... IN/LIGHT OUT
11. Yaw Damper .... AS REQUIRED
12. Autopilot .... OFF (LIGHT OUT)
13. Landing Lights .... AS REQUIRED

HYDRAULICS

UNWANTED OPERATION OF STANDBY PUMP

ST PUMP

1. Hydraulic Standby Pump Switch .... OFF

LOSS OF NO. 1 HYDRAULIC SYSTEM

PMP 1 **PMP 2** AND POSSIBLY **Q UNIT** lights

- Reduce airspeed down to 260 KIAS or MI = 0.76.
- Descend to an altitude not to exceed 45,000 ft. (for airplanes incorporating SB F50-163, maximum altitude extended to 49,000 ft).
- ST BY PUMP Switch ....ON

EFFECTS

LOSS OF	REMARKS
Servo-actuator barrel No. 1	Barrel No. 2 available.
7 Pitch and roll ARTHUR units.	If **Q UNIT** light on, disengage autopilot.
Normal Slat control system	Use EMERG SLATS switch: – Landing in S + FLAPS 48° configuration: • Speed, VREF + 5 kt. • Increase the S+48 landing distance by 180ft/55m and the landing field length by 300 ft / 91 m. – Landing in S + FLAPS 20° configuration: • Speed, VREF + 10 kt. •Increase the S+20 landing distance by 180ft/55m and the landing field length by 300 ft / 91 m.
Normal and EMERG–GEAR: PULL controls.	Free fall extension of the gear. TAB 11
Braking supplied by #1.	Select #2 for braking (landing without antiskid) Additional increase of the determined landing distance or landing field length of: – 25 percent for S + FLAPS 48° landing. – 30 percent for S + FLAPS 20° landing.
Thrust Reverser	Dependent on Thrust Reverser availability.

DEPRESSURIZATION OF HYDRAULIC RESERVOIRS

TKP1 AND/ OR **TKP2**

1. Hydraulic Fluid Level and Pressure .... CHECKED

If pressure starts fluctuating:

2. Conditions Permitting, Descend....................... BELOW 20,000 FEET

LOSS OF NO. 2 HYDRAULIC SYSTEM

PMP 3

•Reduce airspeed down to 260 KIAS or MI = 0.76 max.

If reservoir 2 level is within the red range:

1. ST BY PUMP switch .... OFF

As long as the altitude is above 45,000 ft: (for airplanes incorporating SB F50–163, maximum altitude extended to 49,000 ft).

2. ST BY PUMP Switch .... OFF

EFFECTS

LOSS OF	REMARKS
Servo-actuator barrel No. 2.	Barrel No. 1 available.
Airbrake system.	The landing distance is increased by 600ft/183m (which makes 1,000ft/305m more on the landing field length)
Emergency slat control system.	Use normal slat control system.
No. 2 braking system.	The Park Brake system can still operate with accumulator pressure.
Nose wheel steering system	Use #1 braking system and differential brake pressure.
Flap system. Flap setting	0°Landing with slats only, at VREF + 20 kt: – The S + FLAPS 20° landing distance is increased by 2,000ft/610m (which makes 3,300 ft/1,006m more on the landing field length). This figure includes the consequence of airbrake system loss.
	20° Land using VREF + 5kt.
	48° Land using VREF.

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LOSS OF NO. 3 ENGINE DRIVEN PUMP

PMP 3

NOTE

Avoid using the stand-by pump before initiating descent.

If the stand-by pump is used:

1. ST BY PUMP Switch .... ON
2. No. 2 System Pressure (1,500 – 2,150 psi)........................CHECKED
3. No. 2 Fluid Quantity Indicator.......................................MONITORED

If the stand-by pump is not used:

- Reduce airspeed down to 260 KIAS or MI = 0.76 max.

FLIGHT CONTROLS

FLAP SYSTEM JAMMING OR ASYMMETRY

FLAP ASYM [MAY BE ON]

WARNING

DO NOT CHANGE FLAP LEVER POSITION.

With flaps extended between 0° and 20°:

1. Approach Speed...VREF + 20 KNOTS
2. Landing Distance/Landing Field Length..........ADD 1400/2300 FEET

With flaps extended between 20° and 48°:

1. Approach Speed...VREF + 5 KNOTS
2. Landing Distance/Landing Field Length................ S+20 FLAPS data

CAUTION

If the flap handle has not been selected to 48°, the "landing gear not extended" warning horn may not sound if the landing gear is not properly extended.

8

AILERON SYSTEM JAMMING

CAUTION

Use EMERG AILERON with caution due to possible reverse roll (depending on the location of the jamming).

1. Rudder pedals ....STOP THE ROLL
2. EMERG AILERON trim........................ BANK / TRIM THE AIRPLANE
3. Maximum bank angle ....15°

For Landing:

1. Recommended maximum crosswind .... 14 KTS
2. Use rudder pedals to roll the airplane.
3. Slats-flaps..S + FLAPS 48°
4. Approach speed (zero wind).................................. V_{REF} + 10 KNOTS
5. Fly a shallow final approach.
6. Increase the landing distance by 14% for every 10 KNOTS extra speed.

FSI NOTE

This procedure is found in the Operating Manual, Book 1 of the Abnormal Section (Not in AFM).

RUDDER JAMMING

1. Use differential thrust to control the airplane.
2. Recommended maximum crosswind for landing................... 10 KTS
3. In case of go-around with one engine inoperative, bank the airplane up to 5° to maintain heading.

8

FSI NOTE

This procedure is found in the Operating Manual, Book 1 of the Abnormal Section (Not in AFM).

AIRBRAKES DO NOT RETRACT

AIRBRAKE

With the Airbrakes Extended to Position #1:

Approach Speed:

Slats + Flaps 20° .... VREF + 15 KNOTS

Slats + Flaps 48° .... VREF + 10 KNOTS

Landing Distance/Landing Field Length........................ ADD 480/800 FEET

With Airbrakes Extended to Position #2:

Approach Speed:

Slats + Flaps 20° .... VREF + 20 KNOTS

Slats + Flaps 48° .... VREF + 15 KNOTS

Landing Distance/Landing Field Length........................ ADD 480/800 FEET

NOTE

Place the airbrake control handle in the position which matches the airbrake configuration.

SLAT MONITORING SYSTEM MALFUNCTION

AUTO-SLATS

AT TAKEOFF

No arming of the Automatic Slat Extension System

(Airplanes equipped with automatic monitoring of the flight/ground proximity switches, FSo-166 Service Bulletin)

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1. Airspeed Range .... 1.3 VS to 270 KIAS

AUTO-SLATS

IN FLIGHT

1. Limit Airspeed to Below.. 270 KIAS

SLAT SYSTEM MALFUNCTION

Light on and possible roll.

Cruise

1. Limit Airspeed to Below .... 200 KIAS
2. Autopilot .... DISENGAGED

Approach

1. Slat-Flap Handle Left In .... SLATS
2. Emergency Slats .... ON

If the green light comes on (red light out):

3. Slat-Flap Handle .... S + FLAPS 48
4. Normal Approach At .... VREF

If the green light does not come on:

5. Slat-Flap Handle .... CLEAN

Green Light On	Green Light Out
(Outboard slats are extended) 1. Slat-Flap Handle .... AS NECESSARY (The red light comes on, the green light goes out) 2. Approach speed: • Slats + Flaps 48° VREF + 5 KNOTS • Slats + Flaps 20° VREF + 10 KNOTS **NOTE** Increase landing distance by 180 ft / 55 m and landing field length by 300 ft / 91 m.	(Outboard slats are not extended) 1. Do not extend flaps. **NOTE** Approach speed VREF + 30 KNOTS

9

CAUTION

Do not return EMERG SLATS switch to OFF.

To land in CLEAN configuration:

1. Check AFM (P.5.50.8) whether speed VREF + 30 knots is less than VMBE.
   - In that case, landing distance is increased by 3,000 ft/914 m and the landing field length by 5,000 ft/1,524 m.
   - If not, braking cannot be initiated until the speed has decreased below VMBE: Add to the above figure an extra landing distance of 280 ft/85m (470 ft/143m for landing field length) for each knot of difference between VREF + 30 knots and VMBE.

Q UNIT

Q UNIT

1. Limit Airspeed.. 260 KIAS/0.76 M MAXIMUM
2. Fasten Seat Belts Sign ....ON
3. Autopilot ....DISENGAGE

CAUTION

The pitch and roll control forces may be higher or lower than normal depending on whether the ARTHUR units fails in “high” or “low” speed position.

With Light Control Forces:

Avoid large displacements and rapid movements of the control surfaces.

CAUTION

Maintain speed below 260 KIAS or MI 0.76.

With Heavy Control Forces:

Use normal or emergency trim system and execute a shallow approach at VREF.

NOTE

- The Q UNIT light can illuminate for an ARTHUR Q failure, or #1 hydraulic system inoperative, or in the event of an engine No. 1 PT2 TT2 probe malfunction in icing conditions.
- A No. 1 ADC failure may illuminate the Q UNIT light without incurring any Q-Unit failure.

10

INOPERATIVE ELEVATOR LANDING

Approach Speed:

Slats + Flaps 20° .... VREF + 15 KNOTS

Slats + Flaps 48° .... VREF + 10 KNOTS

Landing Distance/Landing Field Length.................... ADD 1800/3000 FEET

Use very short actuation signals to set stabilizer to desired position.

Make a shallow final approach.

INOPERATIVE STABILIZER LANDING

1. Autopilot .... DISENGAGE

If the stabilizer is jammed in the range +1º to -5º:

Make a shallow approach.

2. Slat-Flap Handle .... S + FLAPS 20°
3. Airspeed .... VREF + 20 KNOTS
4. Landing Distance/Landing Field Length .... ADD 480/800 FEET

CAUTION

Do not reduce engine thrust before touchdown.

The “Landing Gear Not Extended” warning horn may not sound.

If the stabilizer is jammed in the range -5º to -11º:

Make a normal approach.

2. Slat-Flap Handle .... S + FLAPS 48°
3. Airspeed .... VREF

10

LANDING GEAR STEERING BRAKES

LANDING GEAR EXTENSION MALFUNCTION

WARNING

- ONE OR MORE GREEN GEAR LIGHT OUT.
- LANDING GEAR HANDLE LIGHT BLINKING.
- LANDING GEAR NOT EXTENDED WARNING HORN MAY SOUND.

1. Landing Gear Handle.. DOWN - CHECKED
2. Emerg Gear Handle .... PULLED

If three green gear lights come on and the landing gear handle light goes out, the landing gear is down and locked. Do not actuate landing gear controls.

If at least one green light does not come on and the landing gear handle light remains blinking, apply the FREE FALL EXTENSION procedure.

Free Fall Procedure

NOTE

Free fall extension of all 3 gears takes approximately 2 minutes.

3. Airspeed.. NOT LESS THAN 160 KIAS

If necessary, extend the main gear first; one after the other:

4. Manual Main Gear Release Handles .... PULL

   Slip the aircraft left and right while accelerating to 190 KIAS until illumination of each green light is achieved. Gently come back to neutral rudder.

If necessary, then extend the nose gear:

5. Manual Nose Gear Release Handle.. PULL

   Accelerate until illumination of the green light is achieved.

CAUTION

11 Do not actuate landing gear controls once all three gears are locked down. Keep the landing gear down.

Continued on Next Page

LANDING GEAR RETRACTION MALFUNCTION

1. Airspeed..190 KIAS MAX

   In icing conditions or if the takeoff was made through snow or slush on the runway, if the red landing gear lights fail to go out upon retraction of the landing gear, ice may be preventing the main landing gear from locking in the UP position. Cycle the GEAR DOWN and up to get rid of the ice.

   If Non-Icing conditions or if takeoff was made without snow or slush on the runway: Extend and keep the landing gear down.

NOSEWHEEL STEERING FAILURE

1. Release or return the steering control to the neutral position.
2. Use differential braking.

NOSEWHEEL SHIMMY

1. Hold the nose wheel steering control depressed.

NO. 1 BRAKE SYSTEM OR ANTISKID INOPERATIVE

If the Antiskid Test was abnormal:

1. Brake Selector Switch .... #2 OFF

   Perform several successive brake applications.

   Increase the landing distance/landing field length:

   - For landing S + 48° Flaps .... ADD 25%
   - For landing S + 20° Flaps .... ADD 30%

CAUTION

Cautionsly modulate brake prssure to prevent locking of the wheels and thus avoid blow-out of the tires.

NO. 1 AND NO. 2 BRAKE SYSTEMS INOPERATIVE

The thrust reverser and the parking brake are sufficient to permit the aircraft to be brought to rest.

Apply brake pressure cautiously, pulling the parking brake handle to intermediate detent only, and avoid alternate application/release actions 11

NOTE

With **#2P BK** light blinking, residual pressure allows for one brake application only.

FUEL

LOW BOOSTER PUMP PRESSURE

FUEL 1 OR **FUEL 2** OR **FUEL 3**

1. Associated Crossfeed.. OPEN
2. Associated Booster Pump.. OFF

If the light goes out:

Continue flight maintaining balanced fuel levels in wing tanks.

If the light remains on:

3. Crossfeed ....CLOSED
4. Associated Total Fuel Quantity Indicator .... MONITOR

If a fuel loss is evident:

5. Power Lever...CUTOFF
6. Associated Engine Fire Handle... PULL
7. Booster Pump.. OFF
8. Generator Switch... OFF

NOTE

In icing conditions, if the #1 or #2 engine has been shutdown, continue to operate respective engine anti-ice system.

9. Engine Anti-ice Switch.. OFF
10. HP Bleed Switch... OFF
11. PRV Switch (With #2 Engine Inoperative).................................. OFF

If engine #3 is shutdown:

12. Bus Tie Switch... TIED
13. Hydraulic Standby Pump Switch....................... ON (AS REQUIRED)
14. Continue flight maintaining balanced fuel levels in wing tanks.

OR

One Engine Inoperative Approach/Landing/Go-Around Procedures:

- Yellow Tab 6, page A-12

Approach And Landing With Two Engines Inoperative:

- Red Tab 10, pages E-24 and E-25

FUEL TRANSFER SYSTEM INOPERATIVE

XFR

12

1. Associated Transfer Pump.. OFF
2. Fuel Levels.. MONITOR

If feeder tank level is maintaining regulation:

3. Continue flight checking fuel quantity and fuel burn.

If feeder tank level is low, or if fuel loss is evident:

4. Fuel Transfer Shutoff Valve On Affected Side........................ CLOSE
5. Transfer Intercom On Affected Side .... OPEN

FUELING LIGHT ON INFLIGHT

To prevent overboard discharge of fuel through vent outlets:

1. Avoid rapid changes in attitude.
2. Restrict pitch and bank to low angles.
3. Abort flight if conditions permit.

WING TANK LEVEL ABNORMALLY LOW

1. Associated "Low" Tank Transfer Intercom................................ OPEN
2. Associated "Low" Tank Transfer Pump .... OFF
3. Total Fuel Quantity Indicator... MONITOR

If the wing tank level keeps decreasing:

4. Both Transfer Intercoms .... OPEN
5. Transfer Pump (Low Tank) ....ON
6. Transfer Pumps (Associated with Normal Level Tanks).... BOTH OFF

   When the fuel in the low tank is exhausted:
7. Transfer Pumps (Associated with Normal Level Tanks)...... BOTH ON
8. Transfer Pump (Associated with Empty Tank) .... OFF
9. Transfer Intercom Associated With Normal Level Tank .... CLOSE

If wing tank level stops decreasing:

10. Fuel Quantity And Fuel Burn... MONITOR

    If a fuel leak is evident:
11. Engine Shutdown (Yellow Tab 2).....................................COMPLETE

    Maintain balanced fuel in wing tanks.

CAUTION

Fuel crossfeeds must not be used for fuel level balancing unless it is evident the fuel asymmetry is not due to a fuel leak.

FUEL FEEDER TANK LEVEL LOW

LO FUEL

Indication: Rear quantity indicator(s) below green arc and light possibly on.

- **Verify the level drop is not due to a fuel leak.**

  1. TOT Quantity Indications (All 3) .... CROSS-CHECKED
  2. Fuel Quantity Indicators...REAR

- **If a fuel leak is evidenced:**

  1. X FEED associated with the "low" feeder................................ OPEN
  2. BOOSTER associated with the "low" feeder.............................. OFF
  3. Continue flight maintaining balanced fuel levels in wing tanks.

- **If the total air temperature is very low, the failure may be due to fuel freezing.**

In case of extended cruise in cold atmosphere with total air temperature below fuel freezing point, wing-to-feeder tank transfer may be lost. Therefore, monitor the total air temperature, the fuel temperature indicator (SB 136), and carefully monitor the transfer. If necessary, increase mach number or decrease altitude to raise the total air temperature.

- **AIRPLANES EQUIPPED WITH AN EMERGENCY WING-TO-SIDE FEEDER TANK TRANSFER SYSTEM "EMERG FUEL TRANSFER" (AMD-BA F50-175 SB APPLIED)**

Determine feeder tank level:

A Side Feeder Level(s) Low

OR

If Side Feeder Level(s) Low:
- Consult procedure A

If Center Feeder Level Low, with at least one of the side regulations normal:
- Consult procedure B

If Center Feeder Level Low with abnormal side feeder regulation:
- Consult procedure C

1. Emergency Fuel Transfer Switch .... ON AFFECTED FEEDER SIDE
   - Hold the switch in position until feeder level has risen to 700 lbs (320 kgs). Then check **LO FUEL** light goes out.
   - Fold down the switch guard.
2. Keep checking the feeder level. It should go down to 600 lbs (272 kgs) and stabilize. If not, resume the procedure at step A.

B Center Feeder Level Low, with at least one of the side regulations normal

1. X Feed (On the side where feeder regulation is normal)........... OPEN
2. Center Feeder Booster .... OFF
3. XFR Intercom
   (On the side where X Feed is open)........................... AS REQUIRED

13 Continue flight maintaining balanced fuel levels in wing tanks.

Continued on Next Page

(C) Center Feeder Level Low, with abnormal side feeder regulations

1. One of the X Feeds .... OPEN
2. Center Feeder Booster .... OFF
3. Emergency Fuel Transfer Switch .... ON THE SIDE OF THE OPEN X FEED

- Hold the switch in position until affected side feeder level has risen to 800 lbs (363 kgs).
- Fold down the guard.

4. Closely monitor that side feeder. Level should drop to 600 lbs (272 kgs) then stabilize. If not, resume the procedure at step (C).
5. XFR Intercom (On the side where X Feed is open) .... AS REQUIRED

• AIRPLANES NOT EQUIPPED WITH AN EMERGENCY WING-TO-SIDE FEEDER TANK TRANSFER SYSTEM "EMERG FUEL TRANSFER" (AMD-BA F50-175 SB NOT APPLIED)

OR

If Side Feeder Tank Level Low:

• Consult procedure (D)

If two Side Feeder Tanks Level(s) Low:

• Consult procedure (E)

If Center Feeder Level Low, with at least one of the side feeder regulations normal:

• Consult procedure (F)

(D) Side Feeder Tank Level Low

1. Associated X Feed .... OPEN
2. Booster Associated with "LOW TANK" .... OFF
3. XFR Intercom .... AS REQUIRED

Continue flight maintaining balanced fuel levels in wing tanks.

(E) Two Side Feeder Tanks Level(s) Low

1. X Feed (All 2) .... OPEN
2. Side Feeders Booster .... OFF
3. XFR Intercom (All 2) .... OPEN

Continue flight checking tank levels.

(F) Center Feeder Level Low, with at least one of the side feeder regulations normal

1. X Feed (On the Side Where Feeder Regulation is Normal) .... OPEN
2. Center Feeder Booster .... OFF
3. XFR Intercom (On the side where X feed is open) .... AS REQUIRED
4. Apply the same procedure for the other side feeder.

Continue flight maintaining balanced fuel levels in wing tanks.

FEEDER TANK LEVEL HIGH

1. Transfer Pumps (All 3) .... OFF
2. Both Crossfeed Switches .... OPEN
3. “Normal” Level Boost Pump(s) .... OFF

If fuel level keeps increasing in affected feeder tank:

4. Associated Transfer Shutoff Valve .... AS REQUIRED

If fuel level is restored to normal in affected feeder tank:

5. Booster Pumps (All) .... ON
6. X Feed (Both) .... CLOSED
7. XFR Pump (Associated With the Two
   Normal Level Feeder Tanks) .... ON
8. Fuel Management .... AS REQUIRED

ELECTRICAL

TWO GENERATORS INOPERATIVE

GEN AND **GEN**, AND POSSIBLY TRIPPED SWITCHES

1. Associated Generator(s); Volts/Amps .... CHECKED
2. Batteries: Volts/Amps .... CHECKED
3. Associated Generator Switches .... ATTEMPT RESETS (MAX 2 EACH)

If resetting cannot be achieved:

4. Associated Generator Switch(es) .... OFF
5. Bus Tie Switch .... TIED

CAUTION

Switch services off as necessary and isolate **C** and **D** buses to limit load of the operative generator.

BATTERY FAILURE

BAT 1 AND/OR **BAT 2** AND

Associated BAT Switch is tripped.

1. Associated Battery Switch .... ATTEMPT RESET (2 MAX)

BATTERY OVERHEAT

HOT BAT AND AND

1. Associated Battery Switch .... OFF

If battery temperature keeps increasing:

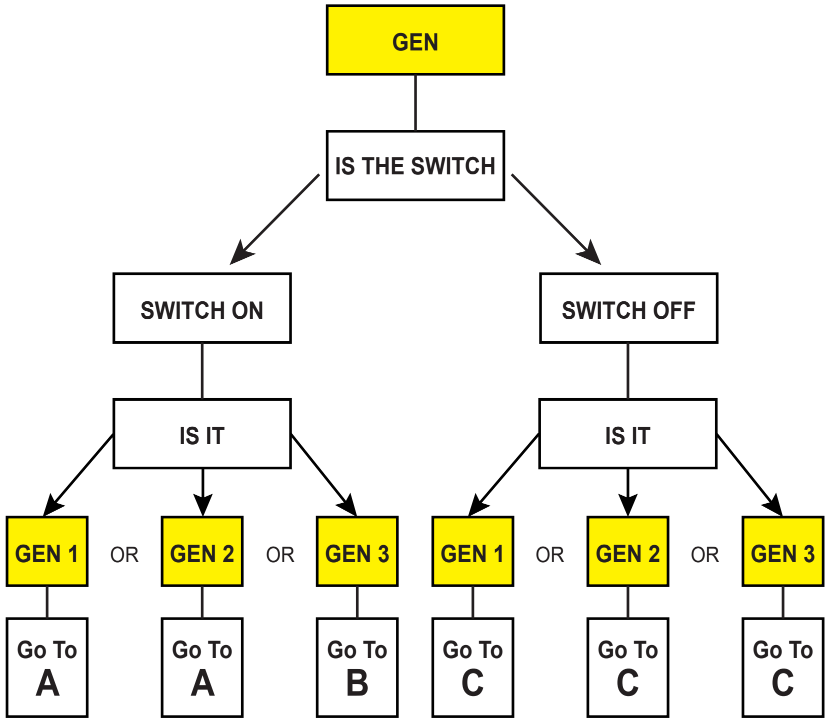
2. Land as soon as possible.

NOTE

If necessary, the faulty battery may be switched back on for landing, providing the **HOT BAT** light has gone out.

ONE GENERATOR INOPERATIVE

GEN 1 OR **GEN 2** OR **GEN 3**



(A) GEN 1 or GEN 2, switch has not tripped:

OR

If voltage is greater than 28.5:

2. On Line Generator Switch............OFF
3. Volts/Amps........... CHECKED

If the voltage is 28.5:

Leave the faulty generator switch off.

Limit the load on the operative generators.

If voltage is normal (28.5):

2. Assoc. Generator Switch.....................ATTEMPT RESET (2 MAX)

If the reset was unsuccessful:

3. Assoc. Generator Switch..............................OFF

Limit the load on the operative generators.

(B) GEN 3, switch has not tripped:

1. Volts and Amps...CHECKED
2. Generator Switch....................................ATTEMPT RESET (2 MAX)

If reset has been unsuccessful:

3. Generator Switch.. OFF
4. Bus Tie Switch.. TIED

Limit the load on the operative generators.

Continued on Next Page

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(C) GEN 1, GEN 2, or GEN 3 switch has tripped:

CAUTION

Only one reset attempt should be made, and only if the associated battery circuit indicates normal loading, and the idle power requirement does not affect the safety of flight.

1. Volts/Amps .... CHECKED
2. Associated Battery Switch .... CHECKED/ON
3. Bus Tie Switch .... FLIGHT NORMAL
4. Associated Power Lever .... IDLE
5. Idle Power .... ACHIEVED
6. Associated Generator Switch .... ON

If the voltmeter goes to maximum, turn generator switch off.

If unsuccessful:

7. Faulty Generator Switch .... OFF
8. Bus Tied Switch .... TIED IN THE CASE OF GEN 3 FAILURE
9. Limit the load on the operating generators.
10. Associated Power Lever .... AS REQUIRED

    Generator Limitations

    - 1 Minute .... 350 AMPS MAX
    - Below FL 390 .... 300 AMPS MAX
    - Above FL 390 .... 250 AMPS MAX

If successful:

7. Associated Power Lever .... AS REQUIRED

INVERTER FAILURE

AC 1 OR **AC 2**

16

THREE INVERTER SYSTEM

Aircraft NOT incorporating Simplified Power AC Generation (Modification M1703), and with SB 214

1. Associated Inverter Switch .... OFF
2. Standby Inverter .... SELECTED TO FAIL SIDE

Verify associated inverter fail warning light out.

TWO INVERTER SYSTEM

Aircraft WITH Simplified Power AC Generation (Modification M1703)

Inverter No. 1 Failure:

1. Inverter #1 Switch .... OFF

Systems inoperative:

- Copilot RMI
- GPWS (If Installed)
- DFDR (If Installed)
- Left electrical rack blower

Inverter No. 2 Failure:

1. Inverter #2 Switch .... OFF
2. Weather Radar Stabilization Switch .... OFF

Systems Inoperative:

- Pilot RMI
- Weather radar antenna stabilization
- Right electric rack blower
- Glareshield lighting
- CVR (if installed)

FAILURE OF BOTH INVERTERS

AC 1 AND AC 2

16

FOR USE WITH THREE INVERTER SYSTEM ONLY

A/C NOT INCORPORATED WITH MOD 1703

1. AC Bus No.1 and No. 2 Voltage .... CHECKED
2. Inverter No. 1 and No. 2 Switches.................................. BOTH OFF
3. Standby Inverter Switch .... CHECKED (CENTER POSITION)
4. Inverter Switch No. 1 .... ON
5. Inverter No. 1 Fail Light .... CHECKED
6. AC Bus No. 1 Voltage .... CHECKED

If an inverter is recovered;

Comply with Inverter Failure procedure; page A-34.

If the malfunction persists:

7. Inverter No. 1 Switch .... OFF
8. Inverter No. 2 Switch .... ON
9. Inverter No. 2 Fail Light .... CHECKED
10. Inverter No. 2 Voltage .... CHECKED

If no inverter is recovered:

11. Inverter No. 1 and No. 2 Switches.................................. BOTH OFF
12. Standby Inverter Switch .... AC NO. 1
13. AC No. 1 Bus Voltage .... CHECKED

NOTE

The following is inoperative:

- Autopilot and yaw damper
- VOR 2, LOC 2, MKR 2, ADF 2 information
- LRN, FMS information
- Weather radar
- Hydraulic system No. 2 pressure indication
- Right radio rack blower

Copilot's side:

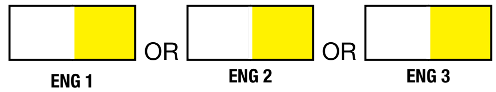
- ADI (EADI)
- HSI (or EHSI)
- Flight Director

Pilot's side:

- HDG No. 2 information on RMI

ANTI-ICE

ENGINE ANTI-ICE SYSTEM INOPERATIVE



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WITH ASSOCIATED ENG ANTI-ICE SWITCH ON

TAT	-30 to –20°C	-20 to –10°C	-10 to –0°C	0 to +10°C
Minimum N1 speed in **cruise** condition	84%	81%	78%	73%
Minimum N1 speed in **approach** condition	78%	78%	78%	73%
One engine inoperative condition	91%	88%	84%	80%

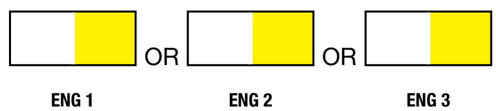
1. N1 Speed Above Minimum Required .... CHECKED

If amber light remains on:

Avoid icing conditions, or leave icing conditions as soon as possible.

Consult Aircraft Operation In Icing Conditions, Additional Information Tab, Pages AI-3 and AI-4 in normal procedures checklist.

ENGINE ANTI-ICE SYSTEM UNWANTED OPERATION



WITH ASSOCIATED ENG ANTI-ICE SWITCH OFF

With TAT greater than +10° C:

If failure affects No. 1 or No. 3 engine:

1. Affected Engine ....REDUCE RPM AS SOON AS POSSIBLE

If failure affects No. 2 Engine:

1. Bleed Air Isolation Switch..ISOLATION
2. Bleed Air HP2 .... OFF
3. No. 2 Engine RPM ....REDUCE RPM AS SOON AS POSSIBLE

AIRFRAME ANTI-ICE SYSTEM INOPERATIVE

AIRFRAME

WITH ASSOCIATED AIRFRAME ANTI-ICE SWITCH ON NORMAL

1. N1 Speed Above minimum required................................CHECKED

If amber light remains on:

2. Airframe Switch ....STANDBY

If amber light remains on:

Avoid icing conditions, or leave icing conditions as soon as possible. 17

Consult *Normal Procedures Checklist*, Aircraft Operation In Icing Conditions, Additional Information Tab, Pages AI-3 and AI-4 in normal procedures checklist.

AIRFRAME ANTI-ICE SYSTEM UNWANTED OPERATION

AIRFRAME

WITH ASSOCIATED AIRFRAME ANTI-ICE SWITCH OFF

With TAT greater than +10° C:

1. Bleed Air Isolation..ISOLATION
2. Bleed Air HP 1 and HP 3 .... OFF
3. Engine No. 1 and
   Engine No. 3 RPM ....REDUCE AS SOON AS POSSIBLE

ICE PROTECTION—LATE ACTIVATION

1. Start Selector Switches (All Three) ....AIRSTART
2. Engine No. 1 and Engine No. 3 Anti-ice Switches ....ON

Wait 30 seconds, then:

3. Engine No. 2 Anti-ice Switch..ON

Wait 30 seconds, then:

4. Airframe Anti-ice Switch ....NORMAL

PRESSURIZATION/AIR CONDITIONING

HIGH CABIN ALTITUDE OR SLOW DEPRESSURIZATION

CABIN

AURAL WARNING

18

Cabin altitude higher than 10,000 feet.

1. Crew Oxygen Masks....DONNED/NORMAL
2. Microphone Selector .... MASK/TESTED
3. Crew Bleed Air Switch....ON or AUTO
4. Cabin Bleed Air Switch....ON or AUTO
5. PRV Switch .... AUTO
6. UP-DOWN Controller....BETWEEN 1 AND 2 O'CLOCK
7. Cabin Pressure Selector Switch ....MANUAL (AS REQUIRED)
8. UP-DOWN Controller.... DOWN (AS REQUIRED)

If necessary:

9. No Smoking Sign....ON
10. Passenger Oxygen Masks....DONNED/CHECKED
11. Nose ....CLOSED
12. Bag Air .... OFF

If necessary:

Execute an Emergency Descent, Emergency Section Tab 5, page E-10, To 14,000 Feet or safe altitude.

HIGH CABIN PRESSURE DIFFERENTIAL

1. Higher Cabin Altitude....SELECTED

If cabin pressure does not decrease:

2. UP-DOWN Controller....BETWEEN 1 AND 2 O'CLOCK
3. Cabin Pressure Selector Switch .... MANUAL
4. UP-DOWN Controller....UP (AS REQUIRED)

If cabin pressure keeps increasing:

5. Bleed Air (Crew and Cabin) .... OFF

Continue flight using the crew and cabin switches to maintain a cabin altitude not higher than 8,000 feet, or a cabin differential pressure not greater than approximately 8.5 psi.

IMPROPER CABIN VERTICAL SPEED

1. UP-DOWN Controller Aligned with Green Mark .... CHECKED
2. Crew Bleed Air Switch .... ON or AUTO
3. Cabin Bleed Air Switch .... ON or AUTO
4. PRV Switch .... AUTO
5. Cabin Rate Knob .... ADJUST CABIN VERTICAL SPEED

If vertical speed cannot be adjusted:

6. UP-DOWN Controller.......................... BETWEEN 1 AND 2 O'CLOCK
7. Cabin Pressure Selector Switch .... MANUAL
8. UP-DOWN Controller.................. ADJUST CABIN VERTICAL SPEED

NO AUTOMATIC PRESENTATION OF PASSENGER OXYGEN MASKS

1. Oxygen Controller.. OVERRIDE
2. Passenger Masks .... DONNED/CHECKED

CAUTION

DO NOT use the reset pushbutton in flight when oxygen system is operating.

CABIN DOOR UNLOCKED

CABIN

NO AURAL WARNING

On the ground only:

Visually check that all 10 access latches are in proper position and that the interior control handle is properly locked.

If these are NOT properly set; do not take off.

In Flight:

CAUTION

Crew and passengers must remain seated, seatbelts fastened and adjusted.

1. FASTEN BELTS light pushbutton.. ON
2. Airspeed.. REDUCE
3. Descend... 10,000' or safe altitude
4. Maneuver .... use Caution
5. Land as soon as possible

NOTE

In case of unpressurized flight conditions, do not pressurize the airplane.

CABIN AIR CONDITIONING OVERHEAT

COND'G OVHT

1. Temperature Controllers .... MANUAL-COLD

If COND'G OVHT stays on:

2. BLEED AIR: CABIN Switch .... OFF

If COND'G OVHT stays on:

3. BLEED AIR: CABIN Switch .... AUTO
4. BLEED AIR: CREW Switch .... OFF

BLEED AIR OVERHEAT

BLEED OVHT

19

1. Bleed Air PRV .... OFF
   a. If BLEED OVHT light starts blinking and then goes out:
      Leave PRV off and continue flight.
   b. If BLEED OVHT light starts blinking and continues to blink:
      Engine No. 2 .... IDLE
   c. If BLEED OVHT light remains on and steady:
      BLEED AIR PRV .... AUTO
2. Successively apply the same procedure for HP1, HP2, and HP3. Should the light blink and continue to blink, the affected thrust should be reduced.

WINDSHIELD

CRACKED WINDSHIELD PANE

1. Airspeed .... 230 KIAS MAXIMUM
2. Cabin Differential Pressure .... DO NOT EXCEED 7.5 PSI
3. Associated Windshield Heat Switch .... NORMAL

WINDSHIELD HEAT SYSTEM MALFUNCTION

XFR

1. Pilot and Copilot Windshield Heat Switches ....SAME POSITION

If possible, before landing:

2. Pilot and Copilot Windshield Heat Switches .... OFF

AUTOPILOT/TRIM

AUTOPILOT FAILURE

AP

AND WITH AIRCRAFT EQUIPPED WITH APS 85 AUTOPILOT + AUDIBLE WARNING

1. Autopilot ....DISENGAGE

AUTOPILOT PITCH TRIM INOPERATIVE

AP TRIM

19

1. Autopilot ....DISENGAGE

OUT OF TRIM CONDITION

MISTRIM

Hold the control wheel firmly.

1. Autopilot ....DISENGAGE

Retrim the aircraft, and attempt to re-engage the autopilot.

MACH TRIM INOPERATIVE

MACH TR

1. Mach Trim Switch ....ON

If mach trim cannot be re-engaged:

Do not exceed 0.78 M without the autopilot engaged.

FLIGHT INSTRUMENTS

PILOT AIR DATA COMPUTER INOPERATIVE

Indication:

- AIR DATA light on, and possibly:
- Q UNIT, AUTO SLATS, and MACH TR lights on
- VMO/MMO warning horn activated
- Pilot altimeter and ALT ALERT flags in view

Airplane equipped with 2 air data computers:

1. Do not use pilot flight director system vertical modes.
2. AIR DATA TRANS Push Switch.................................... ON, AMBER

If AIR DATA light goes out:

Effects:

TABLE A

	LOSS OF	INDICATION	REMARKS
20	Slat Extension prevention at high speed	If AUTO SLATS light is on.	Reduce airspeed to no more than 270 KIAS as soon as possible.
	• Pilot altimeter indication. • ATC 1 transponder altitude report	Pilot altimeter flag in view.	• Use standby or copilot altimeter. • Switch over to ATC 2 transponder.
	Pilot VMO/MMO WARNING:		• The second VMO/MMO warning system remains operative
	• –5° ANU stabilizer travel limit switch.		• Use emergency trim if necessary.
	• Wind and drift on INS. • True airspeed on LRN.	MSG or WRN blinks on LRN.	• Enter speed in LRN manually for DR mode.
	• Mach data to mach trim system.	Possibly, MACH TR light on if MI is higher than 0.78	• Refer to: Mach Trim Inoperative. (Page A-39)

If AIR DATA light stays on:

3. AIR DATA TRANS Push Switch..OUT
4. Autopilot and Yaw Damper.. DISENGAGED
5. Do not exceed MI = 0.78

Continued on Next Page

Airplane equipped with 1 air data computer:

1. AP and YD .... DISENGAGED

   Do not use pilot flight director system vertical modes.

   Do not exceed MI = 0.78.

TABLE B

LOSS OF	INDICATION	REMARKS
ARTHUR unit monitoring.	Possible **Q UNIT** light on.	• Refer to: Arthur Unit Inoperative.
Slat extension prevention at high speed.	If **AUTO SLATS** light is on.	Reduce airspeed to no more 270 KIAS as soon as possible.
ALTITUDE INDICATION: A/C with one ADC • Pilot altimeter indication. • ATC 1 transponder altitude report	Pilot altimeter flag in review.	• Use standby or copilot altimeter. • Switch over to ATC 2 transponder, if copilot has encoding altimeter.
A/C with two ADC • Pilot and copilot altimeters. • ATC 1 and 2 transponder altitude report.	Pilot and copilot altimeters flag in view.	• Use standby altimeter.
• Altitude Monitoring System	ALT ALERT flag in view.	
VMO/MMO WARNING: A/C with one ADC • Pilot system.		• The second warning system remains operative (if SB F50-11 is incorporated
A/C with two ADC • Pilot and copilot systems.		
• 5° ANU stabilizer travel limit switch		• Use emergency trim if necessary
• Outside air temperature indications (SAT/TAT)		• Avoid areas of known icing conditions • Increase fuel consumption in ISA conditions by 5%. • Set engine parameters for cruise at constant Mach No in ISA conditions.
• True airspeed indication. • Wind and drift on INS. • True airspeed on LRN.	**MSG** or **WRN** blinks on LRN.	• Enter speed in LRN manually for DR mode.
• Mach data to mach trim system.	Possibly, **MACH TR** light on if MI is higher than 0.78	• Refer to: Mach Trim Inoperative. (Page A-39)

Continued on Next Page

PILOT AIR DATA COMPUTER INOPERATIVE (Cont)

Aircraft equipped with Mod 1610 incorporated:

Indication:

- **ADC 1** light on, and possibly
- **Q UNIT**, **AUTO SLATS**, and **MACH TR.** lights on
- VMO/MMO warning horn activated
- Pilot altimeter and ALT ALERT flags in view

- Do not use pilot flight director system vertical modes.
- Press the pilot **ADC 1** **ADC 2** pushbutton light:

1. **ADC 2** of Pilot **ADC 1** **ADC 2** Light Illuminates Amber .... CHECKED
2. **XADC** Annunciator on Pilot EADI .... CHECKED

EFFECTS

LOSS OF	INDICATION	REMARKS
Slat Extension prevention at high speed	If **AUTO SLATS** light is on.	Reduce airspeed to no more than 270 KIAS as soon as possible.
Pilot VMO/MMO WARNING:		The second VMO/MMO warning remains operative (copilot).
Horizontal stabilizer deflection at low speeds: deflection limited to -5°		Use emergency trim if necessary.
Mach data to mach trim system.	Possibly, **MACH TR** light on if MI is higher than 0.78	• Refer to: **MACH TRIM INOPERATIVE**.
True airspeed data to corresponding LRN.	**MSG** or **WRN** blinks on LRN.	Manually enter speed in LRN.
Pilot mach airspeed and airspeed bug.	Pilot mach airspeed flag	

Continued on Next Page

Aircraft equipped with Mod 1610 incorporated:

Indication:

- ADC 2 light on, and possibly
- AUTO SLATS lights on
- VMO/MMO warning horn activated
- Copilot altimeter and ALT ALERT flags in view

- Do not use pilot flight director system vertical modes.
- Press the copilot ADC 2 ADC 1 pushbutton light:

1. ADC 1 of Pilot ADC 2 ADC 1 Light Illuminates Amber .... CHECKED
2. XADC Annunciator on Copilot EADI .... CHECKED

EFFECTS

LOSS OF	INDICATION	REMARKS
Slat Extension prevention at high speed	If AUTO SLATS light is on.	Reduce airspeed to no more than 270 KIAS as soon as possible.
Copilot VMO/MMO WARNING:		The first VMO/MMO warning remains operative (pilot).
True airspeed data to corresponding LRN.	**MSG** or **WRN** blinks on LRN.	Manually enter speed in LRN.
Copilot mach airspeed and airspeed bug.	Copilot mach airspeed flag.	
Speed indication on EFIS.		

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PITOT STATIC

PILOT PITOT-STATIC SYSTEM MALFUNCTION

WARNING

Inaccurate airspeed and/or altitude indications.

- **L PITOT** light possibly on.

1. Static Selector Switch .... EMERG

If pilot indications become consistent with copilot's:

- Loss of standby altimeter and rate-of-climb.
- Use other pilot instruments.

If pilot indications remain inaccurate:

- Loss of pilot altimeter, mach airspeed indicator and air data computer.
- Use copilot's indicators.

COPILOT PITOT-STATIC SYSTEM MALFUNCTION

WARNING

21

Inaccurate airspeed and/or altitude indications.

- **Q UNIT** light possibly on.
- **R PITOT** light possibly on.

1. PITOT-STATIC SELECTOR..PANEL ONLY

The selection of the PANEL ONLY position causes the loss of:

- Both Arthur units; consult Q-UNIT inopperative (Yellow TAB 10, A-20)
- Warning horn for landing gear not locked down at speeds below 160 KIAS.
- Cabin differential pressure indications

NOTE

The loss of the copilot pitot-static system causes the yaw damper to disengage.

FLIGHT WITH SUSPECTED BLOCKED PITOT PROBES

WARNING - Frozen or abnormal pilot, copilot and possibly stand-by IAS / MI indications and possibly:

- **AIRSPEED INDICATORS PERFORM LIKE ALTIMETERS** (decreasing in descent and increasing in climb),
- Illumination of one or both following lights: **Q UNIT**, **AUTO SLATS**,
- VMO / MMO audio warning sounds,
- **IAS** miscompare flag on EADI (if any),
- AP disengagement,
- Disagreement with stand-by IAS / MI indications,
- In cruise / level flight: unusual pitch trim activity.

WARNING

INAPPROPRIATE FLIGHT DIRECTOR GUIDANCE MAY BE ACTIVATED. DO NOT FOLLOW CORRESPONDING FD.

CAUTION

Stall aural warning remains reliable.

- Do not apply SLAT MONITORING SYSTEM MALFUNCTION procedure (Yellow Tab 9).

LEVEL FLIGHT

1. Pitch attitude .... Between 2° and 4° nose up
2. Avoid large displacements and rapid movements of control surfaces.

Set N1 as indicated in the table below, corresponding to MI = 0.75 (assumed temperature is ISA -10°C):

Flight Level	Weight	N1	Pitch Attitude
FL 490	24,000 lb	97%	Between 1 and 4 degrees nose up
	22,000 lb	95%	
FL 450	30,000 lb	97%	
	22,000 lb	92%	
FL 410	38,000 lb	98%	
	30,000 lb	93%	
	22,000 lb	91%	
FL 370	38,000 lb	94%	
	30,000 lb	91%	
	22,000 lb	90%	

Continued on Next Page

LEVEL FLIGHT (cont'd)

STATUS (2 BLOCKED PITOT PROBES):

INOPERATIVE / UNRELIABLE ITEMS	OPERATIVE / RELIABLE ITEMS
Basic fight parameters	
IAS / MI / TAS on both ADI and on stand-by instrument.	IRS Ground Speed in FMS CDU. GPS Ground Speed in FMS CDU.
	Pitch and roll attitude on both ADI.
Altitude reportedby XPDR mode C.	Altitude displayed on both altimeters and stand-by instrument (max error +/- 600 ft). GPS altitude in NZ2000 FMS CDU (if any). VS on both VSI.
SAT, ISA deviation	TAT. Temperature data provided by AFIS / Operational flight plan / Weather briefing.
	Heading and Track.
	Wind data provided by AFIS / Operational flight plan / Weather briefing.
Warnings	
VMO / MMO audio warning	
Gear aural warning. Stall aural warning if AUDIO WARN C/Bs pulled	
Flight controls	
Automatic Slats extension.	Stall protection: Automatic Slat Extension; IGN on.
Roll Arthur position inconsistent with actual true flight condition.	Pitch Arthur (Arthur setting dependent on the Pressure (Pt) Measurement by the #1 Engine Probe.
Automatic flight control system	
AP, FD, and YD	
Mach Trim	
Engine	
	Engine primary parameters (N1, ITT, N2, FF) and controls.
Airplane Systems	
	All systems controls and displays

3. After a positive identification of the malfunction, continue the flight while complying with the following procedures for climb and descent phases:

CLIMB

1. AP .... DISENGAGED
2. YD .... DISENGAGED

CAUTION

Do not re-engage AP or YD before pitot probes unblocking.

Continued on Next Page

3. Avoid large displacements and rapid movements of control surfaces.
4. N1 speed ....Climb power as per Performance Manual 4-50
5. Pitch attitude ....Between 4° and 5° nose up

If vertical speed drops below 100 ft / min:

6. Airplane...Level off

If VMO / MMO audio warning sounds:

7. AUDIO WARN A / AUDIO WARN B circuit breakers................. Pulled

CAUTION

All audio warnings (STALL included) are inoperative except TCAS aural warning.

DESCENT

Initiating the descent earlier than scheduled to recover non icing conditions is left to pilot's discretion.

CAUTION

If IAS goes down to 50 kt due to blocked pitot probes, expect loss of airspeed dispaly on both EADI (if any):

- Do not apply PILOT AIR DATA COMPUTER INOPERATIVE procedure.
- Do not apply COPILOT AIR DATA COMPUTER INOPERATIVE procedure.
- Use ADI attitude and the stand-by altimeter unti pitot probes unblocking.

1. AP .... DISENGAGED 21
2. YD .... DISENGAGED

CAUTION

Do not re-engage AP or YD before pitot probes unblocking.

3. Avoid large displacements and rapid movements of control surfaces.
4. Start selector switches (all 3)...AIR START

ANTI-ICE:

5. ENG 1 and ENG 3 ANTI-ICE switches...ON
   - 30 seconds later:
     • ENG 2 ANTI-ICE switch..ON
   - 30 seconds later:
     • AIRFRAME ANTI-ICE switch ....NORMAL
6. N1 speed .... See table below

Continued on Next Page

TAT	Between -30°C and -20°C	Between -20°C and -10°C	Between -10°C and 0°C	0°C and above
N1	84%	81%	78%	73%

7. Airbrake handle.. Position 1
8. Pitch attitude ....Between 0° and 2° nose down
9. Vertical speed indicator ....Between -2,000 and -3,000 ft / min

NOTE

An indicated airspeed increasing in descent is a good evidence of the pitot probes unblocking.

NOTE

Descent should cause airspeed and total air temperature to increase thereby facilitating the pitot probes unblocking and a return to correct IAS indication after about 2 minutes.

NOTE

1. Check airplane altitude frequently on the stand-by altimeter.
2. If prior to the problems, flight was performed at a static temperature lower than the authorized mini mum limit (see 1-15-9), descend as soon as possible until air-data indications are back to normal.

10. After return to unblocked pitot probes situation, wait 1 more minute then:
    - AP and YD...As required

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11. AUDIO WARN A / AUDIO WARN B circuit breakers....... Re-engaged
12. TCAS.. NORMAL - Checked
13. Start selector switches (all 3)...NORMAL
14. Airbrake handle...As required

ANTI-ICE:

- ANTI-ICE ENG 1 and 3 switches ....As required
- ANTI-ICE ENG 2 switch..As required
- ANTI-ICE AIRFRAME switch ....As required

EFIS

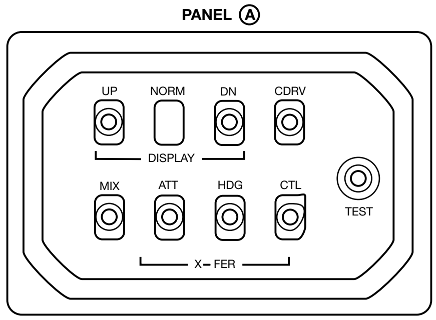
EFIS MALFUNCTIONS

NOTE

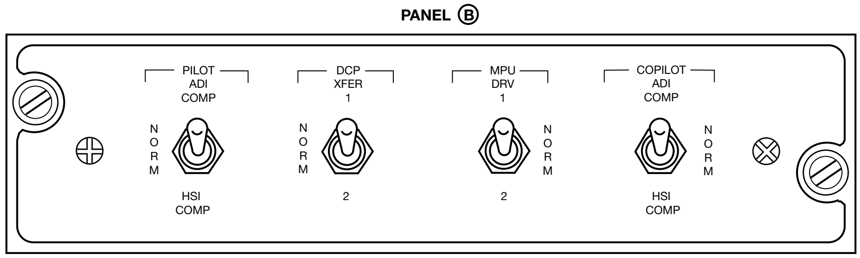
These checklists are applicable to Collins EFIS 85/86 systems.

Various types of avionics equipment may be installed in this aircraft, and specific warning/comparator flags, annunciators, and switches, described herein, may not appear in certain installations. Consult the *Aircraft Flight Manual Supplement* for data relating to specific equipment installation.

PANEL (A)



PANEL (B)



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EADI DISPLAY FAILURE

Indication: Display goes out or color is altered.

1. Display Reversion Switch .... DOWN

EHSI DISPLAY FAILURE

Indication: Display goes out or color is altered.

1. Display Reversion Switch .... UP

SIMULTANEOUS FAILURE OF EADI AND EHSI DISPLAYS ON SAME SIDE

Indication: both displays go out or color is altered, Red FAIL or DRV flag on the EADI and EHSI.

1. Select CDRV or MPU DRV on failed side.
2. AMBER CDRV flag displayed.

If unsuccessful:

3. Select MIX or ADI COMP or HSI COMP.

SUCCESSIVE FAILURE OF EADI AND EHSI DISPLAYS ON SAME SIDE

Indication: Both displays successively go out.

If panel A is installed:

1. Select CDRV on failed side.
2. Select MIX on failed side.

23 If panel B is installed:

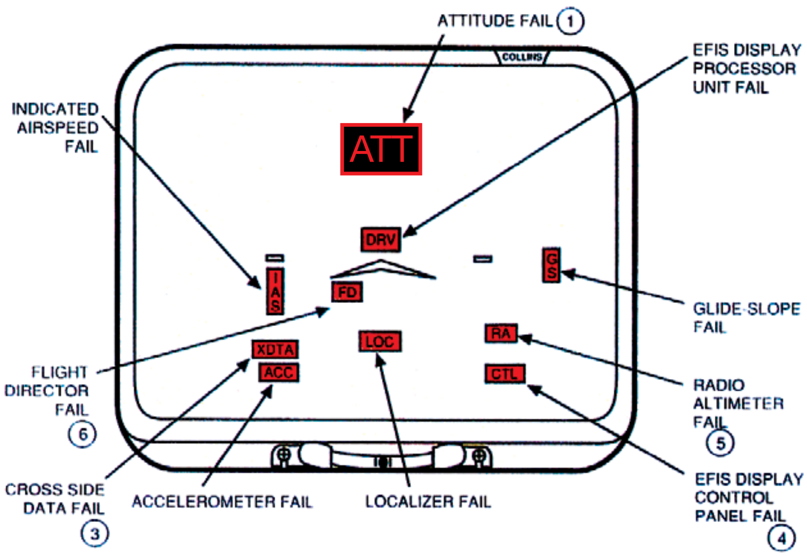
1. Select MPU DRV on the failed side.
2. Select ADI COMP or HSI COMP on failed side.

NOTE

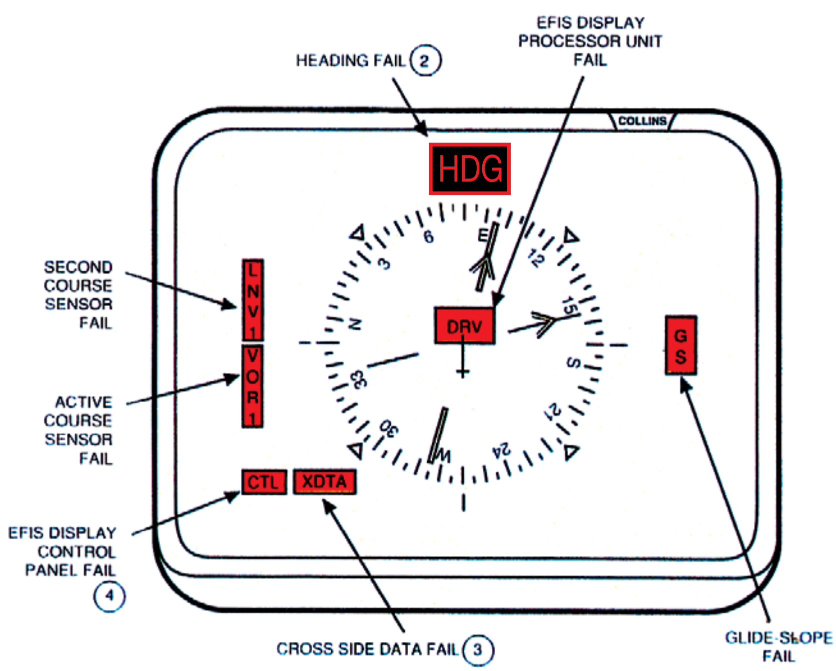
A composite ADI/HSI will be displayed on the Multi-Function Display (MFD).

FAILURE MONITOR FLAGS (ALL RED)

EADI



EHSI

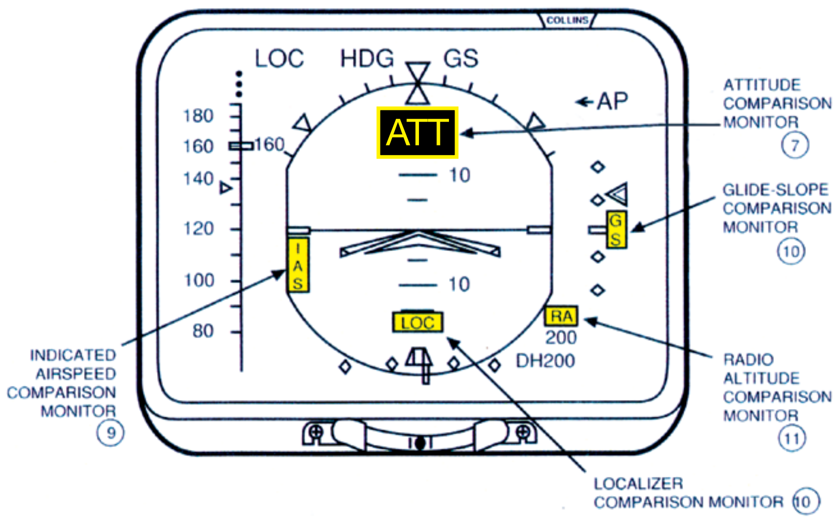


23

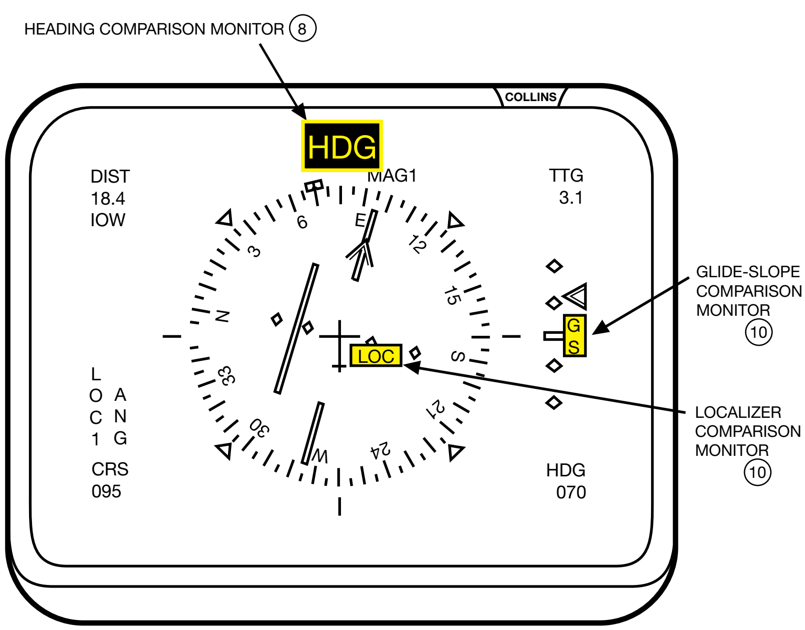
COLLINS EFIS

COMPARISON FLAGS (ALL AMBER)

EADI



EHSI



24

① ATTITUDE FAILURE

Indications: Red ATT flag on EADI
Loss of attitude reference.

1. ATT Transfer Switch .... SELECTED
2. XATT or ATT# Annunciator .... ILLUMINATED

② HEADING FAILURE

Indications: Red HDG flag on EHSI
Displayed heading may be incorrect.

1. HDG Transfer Switch .... SELECTED
2. HDG# or MAG# Annunciator .... ILLUMINATED

③ MPU FAILURE (MULTI-FUNCTION DISPLAY PROCESSOR UNIT)

Indications: MFD Display goes out
Red XDTA flag on both EADIs

CAUTION

- Pilots and copilots HDG bugs are no longer synchronized.
- Comparison monitors are inoperative.

1. MPU DISC or MPU FAIL
   Switch (If Installed)................DEPRESS AND ILLUMINATED AMBER

If red XDTA flag stays on, or if the MPU switch is not installed,

2. MFD Power .... OFF

NOTE

There is no backup for a MPU failure.

24

4 DCP FAILURE (DISPLAY CONTROL PANEL)

Indications: Red CTL or DCP flags on the associated EADI and EHSI

1. CTL XFER or DCP XFER...SELECTED

   XDCP or XCTL annunciators illuminated.

NOTE

- All the displays will be controlled by the operational DCP.
- FD modes are not annunciated on the side with the failed DCP.
- Display brightness remains under control of the failed DCP.

5 RADIO ALTIMETER FAILURE

For aircraft with dual radio-altimeters, consult procedure A, or for aircraft with single radio-altimeter, consult procedure B.

PROCEDURE A.

Indications: Red RA flag on EADI

1. Radio Altimeter
   Transfer Switch....SELECTED TO FUNCTIONAL RADIO ALTIMETER

   Red RA Flag OUT

PROCEDURE B.

Indication: Red RA flag on both EADIs

NOTE

No Backup. Consequences:

- Loss of Radio Altimeter Display
- Loss of DH Warning
- Loss of GPWS (Ground Proximity Warning System)
- Loss of Rising Runway symbol

NOTE

Consult *Aircraft Flight Manual* for possible autopilot minimum altitude limitation with the radio-altimeter inoperative.

25

⑥ FLIGHT DIRECTOR FAILURE

Indication: Red FD fail, command bars removed

If FD #1 fails, and the autopilot is required:

1. AP XFR Switch... SELECTED

NOTE

The autopilot will now follow commands from the copilots flight director.

⑦ ATTITUDE COMPARISON MONITOR

Indications: Amber ATT or PIT or ROL flags on both EADIs, MSTR COMP lights illuminated

1. Autopilot ....DISENGAGE
2. Standby Horizon .... CHECKED
3. Faulty Attitude Source ....IDENTIFY
4. Attitude Transfer Switch (Faulty Side)................................DEPRESS
5. XATT or ATT # Annunciator .... ILLUMINATED
6. Master Comparators..RESET
7. Autopilot .... RE-ENGAGE

⑧ HEADING COMPARISON MONITOR

Indications: Amber HDG flags on both EHSIs MSTR COMP light illuminated

1. Standby Compass .... CHECKED
2. Faulty HDG Source ....IDENTIFY
3. HDG Transfer Switch (Faulty Side) ....DEPRESS
4. XHDG or HDG # Annunciator .... ILLUMINATED
5. Master Comparators..RESET

9 IAS COMPARISON MONITOR

Indication: Amber **IAS** flags on both EADIs

1. Standby Airspeed Indicator..CHECK
2. Faulty Airspeed Indicator..IDENTIFY

10 LOC OR GS COMPARISON MONITOR

Indication: Amber **LOC** or **GS** flags

1. Faulty NAV Source..IDENTIFY
2. Functional NAV Source......................... SELECTED TO BOTH SIDES

NOTE

If operationally feasible, discontinue the approach for any comparison monitor flag.

11 RA COMPARISON MONITOR

*Aircraft with dual radio altimeters only

Indications: Amber **RA** flags on EADIs

1. Faulty Radio Altimeter ....IDENTIFY
2. Radio Altimeter Transfer Switch ....SELECTED TO FUNCTIONAL RADIO ALTIMETER

QUICK REFERENCE

MASTER WARNING INDICATIONS	1
LANDING DISTANCE/LANDING FIELD LENGTH ADDITIVES	2
ELECTRICAL—KEY BUS ITEMS DUAL-FUNCTIONING CIRCUIT BREAKERS GROUND/FLIGHT SWITCHES FUEL CONVERSION CHART	3
28V DC BUS BAR CONSUMERS	4
SINGLE POINT PRESSURE REFUELING (DA-50) SINGLE POINT GRAVITY REFUELING CHECKLIST (DA-50)	5
REFUELING WITH FULL WING TANKS CHECKLIST (DA-50) TO MOVE FUEL FROM THE FEEDER TANKS TO THE WING TANKS (DA-50)	6
SCHEMATICS	7
EMERGENCY/ ABNORMAL PROCEDURE ANNUNCIATORS	

QUICK REFERENCE

MASTER WARNING INDICATIONS

T/O CONFIG. 1

1. Slats not extended.
2. Flaps beyond 22°.
3. Airbrakes not stowed.
4. Stabilizer trim not in takeoff range (–3° to –7°).
5. Autopilot ON.
6. Parking brake (SB 240) ON.

FUELING

1. Bus D - OFF.
2. Gravity Fueling switch ON.
3. Fueling door open.
4. Vent valve handle not stowed.
5. Defueling switch ON.
6. C/B in rear compartment - OUT
7. Any 1 of 3 air vent valves not closed.

APU BLEED

1. APU BLEED Valve is open when it has been commanded to close.

2

LANDING DISTANCE/LANDING FIELD LENGTH ADDITIVES

If landing with Slats + 48 Flaps, use the Slats + 48 Flaps landing charts and apply additives to this landing data. If landing with less than Slats + 48 Flaps, use the Slats + 20 Flaps landing charts and apply additives to this landing data. Antiskid corrections are applied to the basic landing data first before applying all other additives.

Basic Additives

1. No Antiskid 30%/30%
2. No slats and flaps............................VREF +30 ....3000/5000ft
3. Slats only ....VREF +20 ....1400/2300ft
4. S + 20 ....VREF +5 ....charted data
5. Outboard Slats only.........................VREF+25 ....1580/2000ft
6. Outboard Slat +20 ....VREF +10 ....180/300ft
7. No Airbrakes...................................VREF........................600/1000ft

E **Emergency Checklist** A **Abnormal Checklist**

Malfunction..LND DIST/LND FLD LNTH

E Loss of Both Hydraulic Systems, Assumes a Clean Wing, VREF + 30 KT

1. No Antiskid 30%/30%
2. No Slats and Flaps .... 3,000/5,000 FEET
3. No Airbrakes.. 600/1,000 FEET

To land in CLEAN configuration:

1. Check AFM whether speed VREF + 30 kt is less than VMBE.

- In that case, landing distance is increased by 3,000 ft/914m and the landing field length by 5,000 ft/1,524m.
- If not, braking cannot be initiated until the speed has decreased below VMBE: Add to the above figure an extra landing distance of 280 ft/85m (470 ft/143m for landing field length) for each knot of difference between VREF + 30 kt and VMBE.

A Approach and Landing, 2 Engines Inoperative

1. Slats Only, VREF + 20 KT .... 1,400/2,300 FEET
2. S + 20 Flaps, VREF + 10 KT .... 180/300 FEET

A Approach and Landing, One Engine Inoperative

1. S + 20 Flaps, VREF + 5 KT .... CHART/CHART DATA
2. Slats Only, VREF + 20 KT .... 1,400/2,300 DATA

Continued on Next Page

(A) Loss of #1 Hydraulic System

1. No Antiskid:

a. S + 20 Flaps..30%/30%

b. S + 48 Flaps..25%/25%

2. Emergency Slats + 48, VREF + 5 KT .... 180/300 FEET

3. Emergency Slats + 20, VREF + 10 KT .... 180/300 FEET

(A) Loss of #2 Hydraulic System

1. No Airbrakes .... 600/1,000 FEET

2. Slats Only, VREF + 20 KT .... 1,400/2,300 FEET

(A) Airbrakes Do Not Retract

1. With Airbrakes in Position 1:
   a. S + 20 Flaps, VREF + 15 KT .... 480/800 FEET
   b. S + 48 Flaps, VREF + 10 KT .... 480/800 FEET
2. With Airbrakes in Position 2:
   a. S + 20 Flaps, VREF + 20 KT .... 480/800 FEET
   b. S + 48 Flaps, VREF + 15 KT .... 480/800 FEET

(A) Flap Asymmetry

1. S + Up to 20 Flaps, VREF + 20 KT .... 1,400/2,300 FEET
2. S + 20 to 48 Flaps, VREF + 5 KT.................... CHART/CHART DATA

(A) Elevator Inoperative Landing

1. S + 20 Flaps, VREF + 15 KT .... 1,800/3,000 FEET
2. S + 48 Flaps, VREF + 10 KT .... 1,800/3,000 FEET

(A) Stabilizer Inoperative Landing

1. If Stab is +1 to –5 Degrees:
   a. S + 20 Flaps, VREF + 20 KT .... 480/800 FEET
2. If Stab is –5 to –11 Degrees:
   b. S + 48 Flaps, VREF.................................... CHART/CHART DATA

ELECTRICAL—KEY BUS ITEMS

3

BUS A

1. LANDING GEAR CONTROL
2. AIRBRAKE CONTROL
3. NORMAL PITCH TRIM
4. CENTRAL ADC
5. HP 1 & 2 & PRV
6. ANTI-ICE ENGINES 1 & 2
7. FIRE DETECTION ENGS. 1, 2, BAGGAGE & AFT COMPARTMENT
8. #2 BRAKES
9. GAGES ENGINES 1 & 2
10. AUTO PRESSURE CONTROL AND DUMP CIRCUIT
11. STANDBY HYDRAULIC PUMP CONTROL
12. SLATS INDICATOR
13. THRUST REVERSER
14. PILOT'S INVERTER

BUS C

1. NOSE WHEEL STEERING
2. TRIM INDICATOR
3. AILERON TRIM
4. COCKPIT TEMPERATURE CONTROL
5. #1 HYDRAULIC QUANTITY INDICATOR
6. FUEL X-FEED 1-2
7. STBY AIRFRAME ANTI-ICE
8. OIL PRESS/TEMP GAGE #1 AND #2

BUS B

1. EMERGENCY SLATS
2. EMERGENCY PITCH TRIM
3. MACH TRIM
4. ANTISKID
5. ISOLATION VALVE
6. HP 3
7. STANDBY INVERTER
8. FLAP/AIRBRAKE INDICATOR
9. FUEL SYSTEM INTERCONNECT
10. COPILOT'S INVERTER
11. #3 ENG ANTI-ICE
12. FIRE DETECTION—#3 ENG., APU
13. NORMAL AIRFRAME ANTI-ICE
14. GAGES ENGINE 3
15. AUTOPILOT

BUS D

1. FLAPS
2. RUDDER TRIM
3 CABIN TEMPERATURE CONTROL
4. #2 HYDRAULIC QUANTITY INDICATOR
5. FUEL X-FEED 2-3
6. EMERGENCY AILERON TRIM
7. OIL PRESS/TEMP GAGE #3

DUAL-FUNCTIONING CIRCUIT BREAKERS

PRIMARY (LABELED)	SECONDARY
ST-BY PUMP	BRAKE SELECTOR
ENG 2	PRV + H.P. 2
HP BLEED 3	ISOLATION VALVE
RH AUTO SLATS	EMERGENCY SLATS
TRANSFER INTERCOM	EMERGENCY FUEL TRANSFER (SB 175)

GROUND FLIGHT SWITCHES

LEFT MAIN GEAR	NOSE GEAR	RIGHT MAIN GEAR
Switch No. 1	**Switch No. 1**	**Switch No. 1**
Stall No. 1 Test Stall warning flight—left vane Standby hydraulic pump	Landing gear control solenoid	Stall No. 2 Test Stall warning flight-right vane
Switch No. 2	**Switch No. 1 or 2** Nosewheel steering amplifier	**Switch No. 2**
Air-conditioning valve control Air data computer APU ground flight relay Battery blower Cabin pressurization Engine-starting relays Landing gear control solenoid Nose fan control Single-point refueling Standby horizon Takeoff warning Thrust reverser—ground only	**Switch No. 1 and 2** Brake control antiskid Engine No. 2 fail-ground	Air-conditioning valve control Air data computer APU ground/flight relay Battery blower Cabin pressurization Engine starting relays INS Nose fan control Standby horizon Single-point refueling Takeoff warning Thrust reverser—ground only

3

FUEL CONVERSION CHART

JET FUEL—6.75 POUNDS AT 60° F

Gallons	Liters	Weight	Gallons	Liters	Weight	Gallons	Liters	Weight
50	189	338	1050	3974	7088	2050	7759	13838
100	378	675	1100	4163	7425	2100	7948	14175
150	568	1013	1150	4353	7763	2150	8138	14513
200	757	1350	1200	4542	8100	2200	8327	14850
250	946	1688	1250	4731	8438	2250	8516	15188
300	1135	2025	1300	4920	8775	2300	8705	15525
350	1325	2463	1350	5110	9113	2350	8895	15863
400	1514	2700	1400	5300	9450	2400	9084	16200
450	1703	3038	1450	5488	9788	2450	9273	16537
500	1892	3375	1500	5677	10125	2500	9462	16875
550	2082	3713	1550	5867	10463	2550	9652	17212
600	2271	4050	1600	6056	10800	2600	9841	17550
650	2460	4388	1650	6245	11138	2650	10030	17887
700	2650	4725	1700	6434	11475	2700	10219	18225
750	2839	5063	1750	6623	11813	2750	10409	18562
800	3028	5400	1800	6813	12150	2800	10598	18900
850	3217	5738	1850	7002	12488	2850	10787	19237
900	3406	6075	1900	7191	12825	2900	10976	19575
950	3596	6413	1950	7381	13163	2950	11166	19912
1000	3785	6750	2000	7570	13500	3000	11355	20250

4

28V DC PRIMARY BUS A

28V DC AUXILIARY BUS C

OVERHEAD CIRCUIT BREAKER PANEL

BUS BAR A CONSUMERS			BUS BAR C CONSUMERS
DC 1 ADF 1 DME 1 VOR 1 ICS 1	VHF 1 ATC 1	RADIO NAV	INS ADEU RAD ALT WEATHER RADAR TAPE

CENTER CIRCUIT BREAKER PANEL

BUS BAR A CONSUMERS			BUS BAR C CONSUMERS
INST LH LANDING LH ANTICOL FIN CKPIT READING DIGIT 1 NAV	CENTER LH OVERHEAD	LIGHTS MISC	GALLEY CABIN READING CREW SEATS STROBE
INV 1		INVERTER	
IGNITER AUTO REVERSE CONTROL REVERSE WARN ITT 1 N1-N2 1 CPTR 1 IGNTR 1	ITT 2 N1-N2 2 CPTR 2 IGNTR 2	ENGINE	OIL 1 OIL 2
BOOST 1 BOOST 2 FUEL FLOW 1 GAGES LH	FUEL FLOW 2 GAGES CENTER	FUEL	XFR 1 XFR 2
WSHLD FRONT LH LH PITOT HEAT WIPER LH ENGINE 1 COND'G CREW	CABIN PRESS ENGINE 2 HP BLEED DV WINDOW	ANTI-ICE	STD-BY AIR FR AFT SIDE WINDOW CKPT TEMP CONTROL CABIN TEMP NOSE FAN CKPT FLOOR HEATING
LG CONTROL STD-BY PUMP		HYDR	NOSE WHL HYDR 1
A/B CONTROL Q UNIT STAB NORMAL	LH AUTO SLAT SLAT INDIC	FLT CONTROL	A/A INDIC TRIM INDIC TRIM AILERON

RH CIRCUIT BREAKER PANEL

BUS BAR A CONSUMERS			BUS BAR C CONSUMERS
LIGHTS WARN A-B LIGHTS WARN A AUDIO WARN A BAT TEMP REAR CMPT BAG COMPT	ENGINE FAIL 2 DETECT 1 DETECT 2 EXTING 1-2	FIRE WARNING	
AIR DATA AIR DATA AIR DATA EXT TEMP FLIGHT RECORD A/P RUD A/P ELV A/P AIL A/P CMPTR HRZN STD-BY PWR INST 1 ANNUNC LH COMPAR FD 1		AFCS	

DISTRIBUTION OF THE 28V DC BUS BAR A AND C CONSUMERS

28V DC PRIMARY BUS B

28V DC PRIMARY BUS D

OVERHEAD CIRCUIT BREAKER PANEL

BUS BAR B CONSUMERS		BUS BAR D CONSUMERS
OMEGA HF CONTRO HP PWR PUBLIC ADDRESS ICS 2 DG 2	<u>RADIO</u> <u>NAV</u>	IFLIT FONE ADF 2 DME 2 VOR 2 VHF 2 ATC2

CENTER CIRCUIT BREAKER PANEL

INV 2 STD BY	<u>INVERTERS</u>	
CABIN INDIRECT DIGIT 2		INST RH ANTICOLL ELL BELTS NO SMK'G TOILET CKPT RH READING CENTER RH RAZOR CABIN DISPLAY LANDING RH
ITT 3 N1 - N2 3 CMPTR 3 IGNTR 3	<u>ENGINE</u>	OIL 3 SYNCHRO
FUEL FLOW 3 GAGE RH BOOST 3 XFR INTERCOM	<u>FUEL</u>	XFR PUMP 3 PRESSURE FUELING X-FEED 2-1 X-FEED 2-3
LG INDIC ANTISKID	<u>HYDR</u>	HYDR 2
ENGINE 3 HP BLEED 3 WSHLD DEFOG AIR FR COND'G CABIN BAG PRESS	<u>ANTI-ICE</u> <u>COND'G</u>	CAB TEMP CONTROL WSHLD FRONT RH RH PITOT HEAT WIPER RH
RH AUTO SLAT FLAP A/B INDIC STAB EMERG	<u>FLT</u> <u>CONTROL</u>	ROLL EMERG TRIM RUDDER FLAP CONTROL

RH CIRCUIT BREAKER PANEL

LIGHTS WARN B AUDIO WARN B EMERG LIGHTS DETECT 3 EXTING 3 CB PANEL	<u>FIRE</u> <u>WARNING</u>	
	<u>AFCS</u>	INST 2 ANNUNC RH FD 2

DISTRIBUTION OF THE 28V DC BUS BAR B AND D CONSUMERS

SINGLE POINT PRESSURE REFUELING (DA-50)

1. Refueling Panel Door .... OPEN
2. Refueling Coupling Cover .... REMOVE
3. Single Point Refueling .... CONNECT
4. Wing Tank Refueling Switches .... ON
5. Rear Tanks Switch .... HI

NOTE

Assure red “Stop Fueling” light is out and green “Fueling O.K.” light is illuminated.

6. Start Refueling .... PUMPING PRESS. 30–50 PSI
7. Refueling Test Lever .... ROTATE
   (When Fuel Truck Meter Stops, Rotate Lever Back to OFF)
8. Rear Tanks Switch .... LO

NOTE

If full load is desired switch rear switch to HI after fuel truck meter stops.

When Tanks are Filled to Desired Level

1. All Switches .... OFF
2. Refueling Nozzle .... DISCONNECT
3. Refueling Coupling Cover .... INSTALL
4. Vent Valve Control Lever .... DOWN
5. Refueling Panel Door .... CLOSED

SINGLE POINT GRAVITY REFUELING CHECKLIST (DA-50)

NOTE

Unlock and remove the left wing tank filler port cap. Insert fuel nozzle and start pumping. The following steps need to be accomplished in order to move fuel from left wing tank to the center wing, right wing and feeder tanks.

Interior Setup

1. APU or GPU..ON
2. Gravity Refueling Switch ....ON
3. Left Transfer Pump ....ON
4. Left Boost Pump..ON
5. Left Crossfeed .... OPEN

Refueling Panel Setup

6. Defueling Switch..ON
7. Right Wing Tank Refueling Switch..ON

After Refueling Check-Interior

1. Gravity Refueling Switch .... OFF
2. Left Transfer Pump .... OFF
3. Left Boost Pump.. OFF
4. Left Crossfeed ....CLOSED

Refueling Panel Setup

5. Defueling Switch.. OFF
6. Right Wing Tank Refueling Switch.. OFF
7. Refueling Panel Door..CLOSED

REFUELING WITH FULL WING TANKS CHECKLIST (DA-50)

NOTE

These procedures are intended to avert fuel spills on the ramp and a continuous STOP FUELING light on the refueling panel when venting the tanks.

The in flight procedure is for use after a short flight and the center feeder tank has not decreased to regulation level and more fuel needs to be added for the next leg of the flight. This procedure should be performed in-flight.

The On Ground procedure is for the same situation except, the crew is unaware that more fuel is needed for the next flight until the aircraft is on ground and the engines are shut down.

Inflight Procedure

1. Fuel Transfer Intercoms .... BOTH OPEN
2. L & R Wing Transfer Pumps... OFF
3. Operate in this Configuration for 15 Minutes.
4. L & R Wing Transfer Pumps...ON
5. Fuel Transfer Intercoms ....BOTH CLOSED

6

On Ground Procedure

NOTE

Aircraft must have a continuous power source for the on ground procedure. Don’t use battery power for this operation.

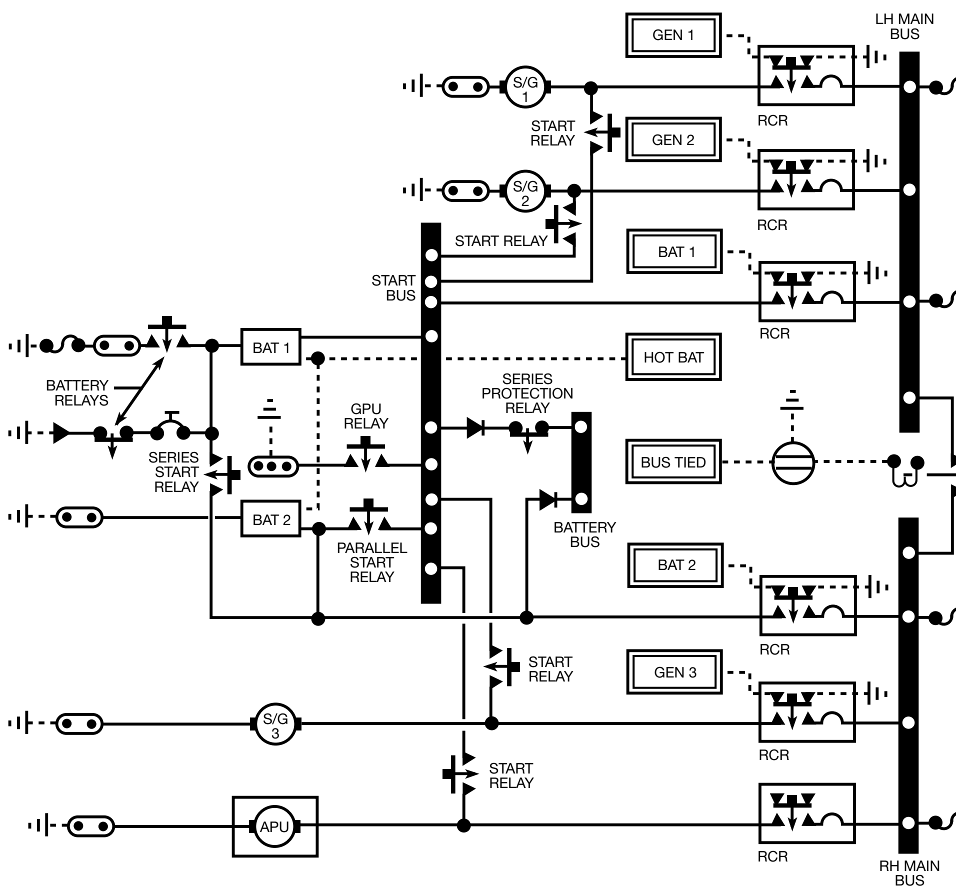
1. Fuel Transfer Intercoms .... BOTH OPEN
2. Center Wing Tank Transfer Pump..ON
3. Emergency Transfer Switch ....LEFT/RIGHT
   (Hold in each position until lateral feeder tank quantity rises 250lb.)
4. Center Wing Tank Pump.. OFF
5. Fuel Transfer Intercoms ....BOTH CLOSED

TO MOVE FUEL FROM THE FEEDER TANKS TO THE WING TANKS (DA-50)

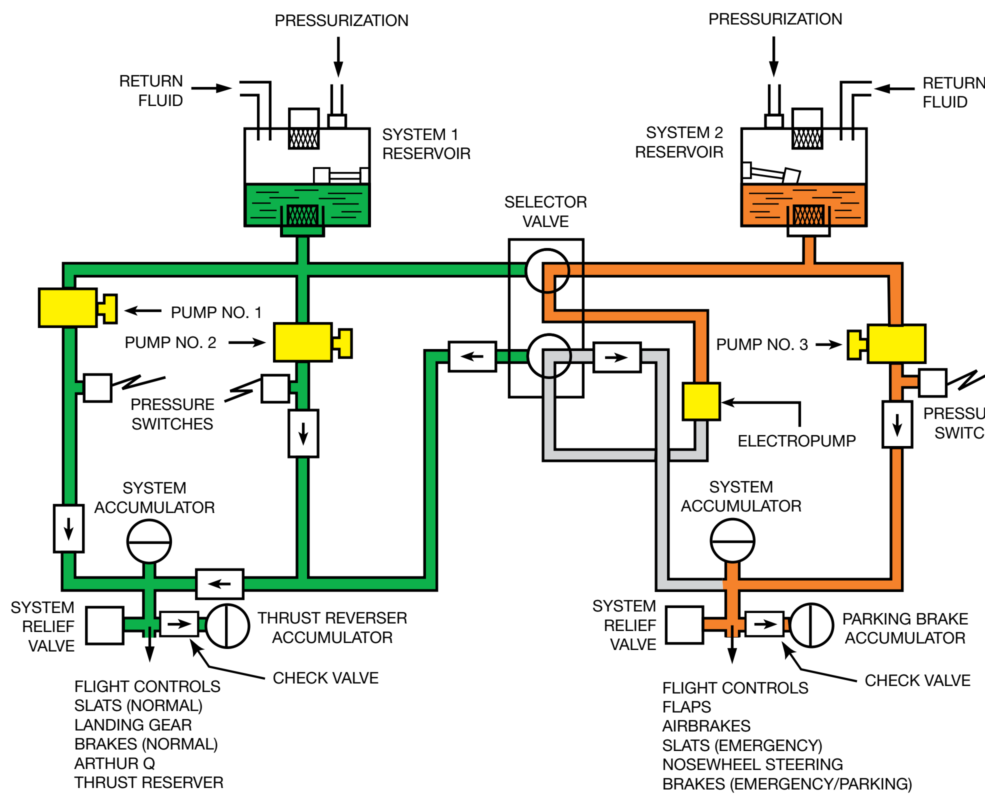
1. APU or GPU.. START
2. Refueling Panel Door.. OPEN
3. Vent Valve Control Lever.. UP
   (Assure green FUELING OK light is illuminated.)
4. Wing Tank Refueling Switches.. ON
5. Defueling Switch.. ON
6. Left & Right Crossfeeds.. OPEN
7. All 3 Boost Pumps.. ON

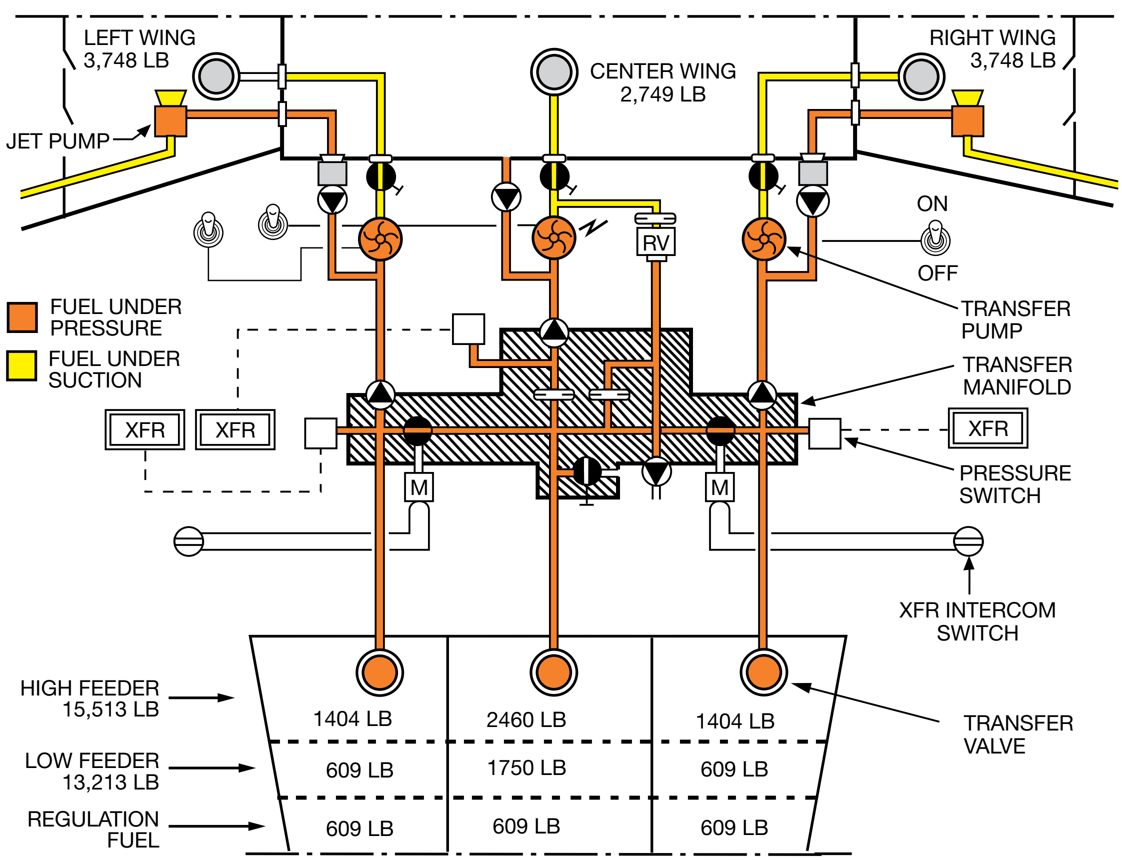
When Finished

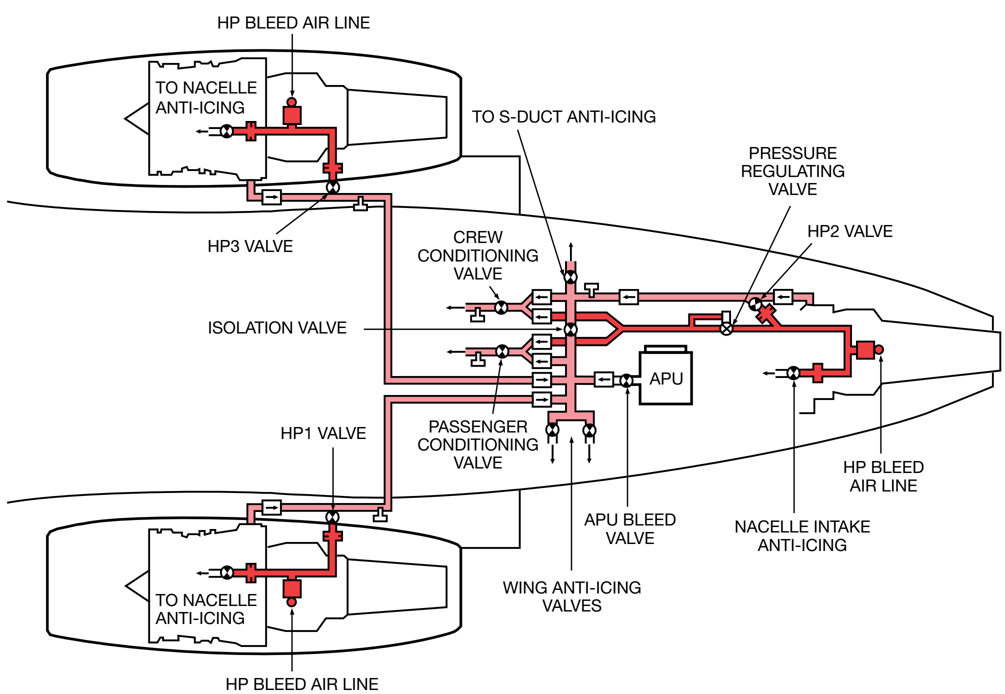
1. All 3 Boost Pumps.. OFF
2. Left & Right Crossfeeds.. CLOSED
3. Defueling Switch.. OFF
4. Wing Tank Refueling Switches.. OFF
5. Vent Valve Control Lever.. DOWN
6. Refueling Panel Door.. CLOSED

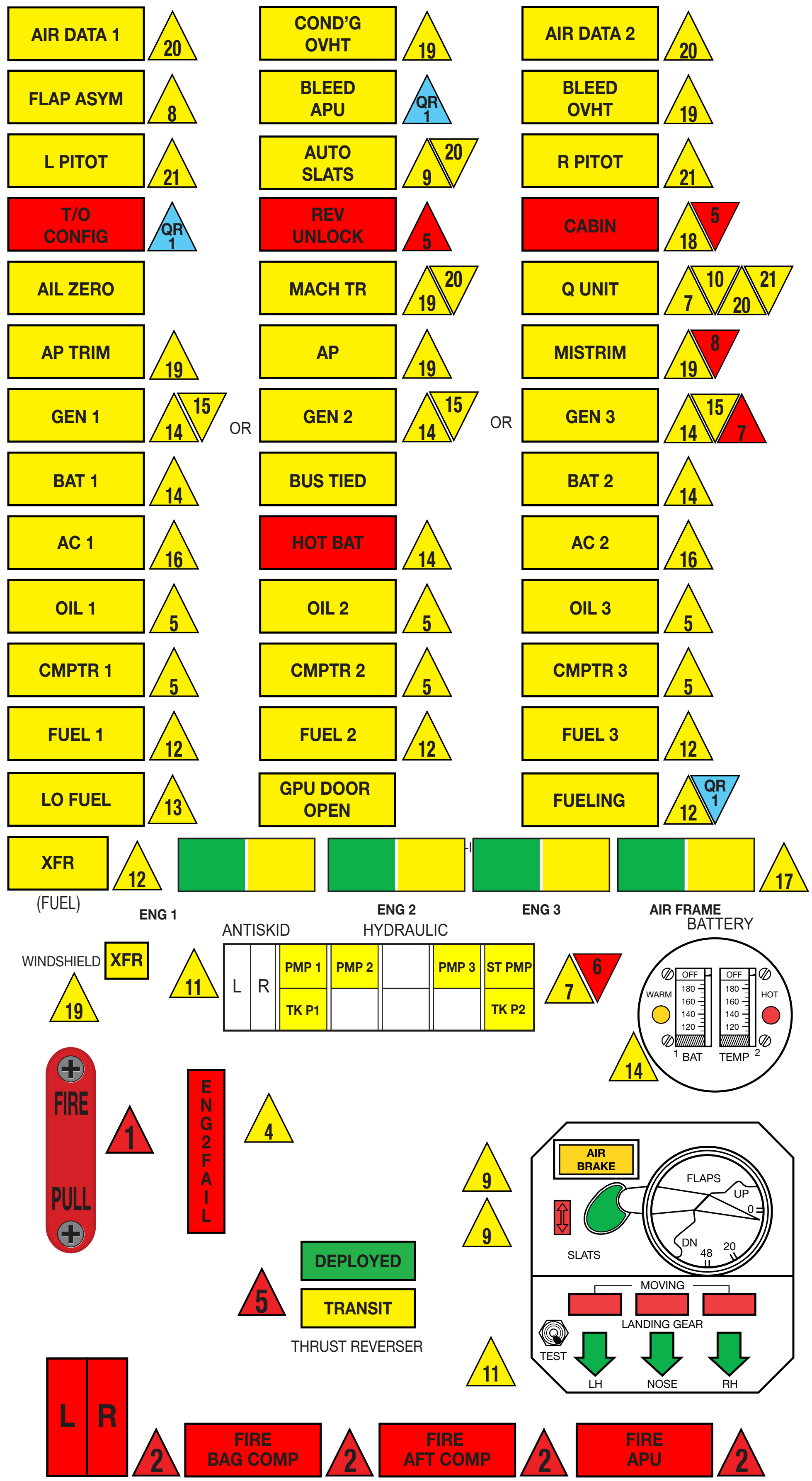

LH MAIN BUS
GEN 1
S/G 1
RCR
START RELAY
GEN 2
S/G 2
RCR
START RELAY
BAT 1
START BUS
RCR
BAT 1
HOT BAT
BATTERY RELAYS
SERIES PROTECTION RELAY
GPU RELAY
BUS TIED
SERIES START RELAY
BAT 2
PARALLEL START RELAY
BATTERY BUS
BAT 2
RCR
START RELAY
GEN 3
S/G 3
RCR
START RELAY
APU
RCR
RH MAIN BUS

FOR TRAINING PURPOSES ONLY




LEFT WING
3,748 LB
CENTER WING
2,749 LB
RIGHT WING
3,748 LB
JET PUMP
RV
ON
OFF
FUEL UNDER PRESSURE
FUEL UNDER SUCTION
TRANSFER PUMP
TRANSFER MANIFOLD
XFR
XFR
XFR
M
M
PRESSURE SWITCH
XFR INTERCOM SWITCH
HIGH FEEDER
15,513 LB
LOW FEEDER
13,213 LB
REGULATION FUEL
1404 LB
2460 LB
1404 LB
609 LB
1750 LB
609 LB
609 LB
609 LB
609 LB
TRANSFER VALVE


HP BLEED AIR LINE
TO NACELLE ANTI-ICING
TO S-DUCT ANTI-ICING
PRESSURE REGULATING VALVE
HP2 VALVE
HP3 VALVE
CREW CONDITIONING VALVE
ISOLATION VALVE
APU
PASSENGER CONDITIONING VALVE
HP1 VALVE
HP BLEED AIR LINE
APU BLEED VALVE
NACELLE INTAKE ANTI-ICING
WING ANTI-ICING VALVES
TO NACELLE ANTI-ICING
HP BLEED AIR LINE


AIR DATA 1 20
COND'G OVHT 19
AIR DATA 2 20
FLAP ASYM 8
BLEED APU QR 1
BLEED OVHT 19
L PITOT 21
AUTO SLATS 9 20
R PITOT 21
T/O CONFIG QR 1
REV UNLOCK 5
CABIN 18 5
AIL ZERO
MACH TR 19 20
Q UNIT 7 10 20 21
AP TRIM 19
AP 19
MISTRIM 19 8
GEN 1 14 15
OR
GEN 2 14 15
OR
GEN 3 14 15 7
BAT 1 14
BUS TIED
BAT 2 14
AC 1 16
HOT BAT 14
AC 2 16
OIL 1 5
OIL 2 5
OIL 3 5
CMPTR 1 5
CMPTR 2 5
CMPTR 3 5
FUEL 1 12
FUEL 2 12
FUEL 3 12
LO FUEL 13
GPU DOOR OPEN
FUELING 12 QR 1
XFR 12
(FUEL)
ENG 1
ENG 2
ENG 3
AIR FRAME
17
ANTISKID
HYDRAULIC
L R
PMP 1
PMP 2
PMP 3
ST PMP
TK P1
TK P2
11
7 6
BATTERY
OFF 180 160 140 120
WARM
HOT
1 BAT TEMP 2
14
WINDSHIELD XFR
19
FIRE PULL
1
ENG 2 FAIL
4
9
9
AIR BRAKE
FLAPS
UP
0
DN
48
20
SLATS
DEPLOYED
TRANSIT
5
THRUST REVERSER
11
MOVING
LANDING GEAR
TEST
LH
NOSE
RH
L R
2
FIRE BAG COMP
2
FIRE AFT COMP
2
FIRE APU
2